

1 Turkey B cell Transcriptome Profile During Turkey
2 Hemorrhagic Enteritis Virus (THEV) Infection Highlights
3 Upregulated Apoptosis and Breakdown Pathways That May
4 Mediate Immunosuppression

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Abstract

Infection with Turkey Hemorrhagic Enteritis Virus (THEV) can cause hemorrhagic enteritis, which affects young turkeys. This disease is characterized by bloody diarrhea and immunosuppression (IMS), which is attributed to apoptosis of infected B cells. Secondary infections due to IMS exacerbates economic losses. We performed the first transcriptomic analysis of a THEV infection to elucidate the mechanisms mediating THEV-induced IMS. After infecting and sequencing mRNAs of a turkey B-cell line, trimmed reads were mapped to the host turkey genome and gene expression was quantified with StringTie. Differential gene expression analysis was followed by functional enrichment analyses using gprofiler2 and DAVID from NCBI. RT-qPCR of select genes was performed to validate the RNA-seq data. A total of 2,343 and 3,295 differentially expressed genes (DEGs) were identified at 12-hpi and 24-hpi, respectively. The DEGs correlated with multiple biological processes including apoptosis, ER unfolded protein response, and cell maintenance. Multiple pro-apoptotic genes, including *APAF1*, *BMF*, *BAK1*, and *FAS* were upregulated. Genes that play a role in ER stress-induced unfolded protein response including *VCP*, *UFD1*, *EDEM1*, and *ATF4* were also upregulated and may contribute to apoptosis. Our data suggest that several biological processes and pathways including apoptosis and ER response to stress are important aspects of the host cell response to THEV infection. It is possible that interplay between multiple processes may mediate apoptosis of infected B-cells, leading to IMS.

Keywords

Turkey hemorrhagic enteritis virus (THEV), Adenovirus, RNA sequencing, Apoptosis, Immunosuppression, ER stress, B cell.

1. Introduction

Turkey hemorrhagic enteritis virus (THEV) belongs to genus *Siadenovirus*, family *Adenoviridae*, and infects turkeys, chickens, and pheasants (1, 2). THEV is transmitted via the fecal-oral route and causes hemorrhagic enteritis (HE) in turkeys, a debilitating disease affecting predominantly 6-12-week-old turkey poults characterized by immunosuppression (IMS), lack of vitality, splenomegaly, intestinal lesions leading to bloody diarrhea, and up to 80% mortality (3–6). The clinical disease usually persists in affected flocks for 7-10 days, causing death and economic losses. However, secondary bacterial infections may extend the duration of illness and increase mortality for an additional 2-3 weeks due to the immunosuppressive nature of the virus, exacerbating the economic losses (5, 7). Naturally occurring low pathogenic (avirulent) strains of THEV have been isolated; these strains cause subclinical infections but retain the immunosuppressive effects. Since its isolation from a pheasant spleen, the Virginia Avirulent Strain (VAS) has been effectively used as a live vaccine despite the immunosuppressive side-effects. However, the vaccinated birds are rendered more susceptible to opportunistic infections and death than unvaccinated cohorts (4, 5, 8–10).

It is well established that THEV primarily infects and replicates in turkey B-cells of the bursa and spleen and to a lesser extent in macrophages, inducing apoptosis and necrosis (6, 8). Consequently, a significant drop in the number of B-cells (specifically, IgM+ B-cells) and macrophages ensues along with increased T-cell counts with abnormal ratios of T-cell subpopulations (CD4+ and CD8+) (6, 8, 11). The cell death seen in the infected B-cells and macrophages is generally proposed as the major cause of THEV-induced IMS as both humoral and cell-mediated immunity are impaired (5, 8). Immunopathogenesis via cytokines from T-cells and macrophages has also been suggested as a mechanism of apoptosis leading to IMS. It is thought that virus replication in the spleen attracts T-cells and peripheral blood macrophages, which results in T-cell activation by cytokines from activated macrophages and vice versa. The activated T-cells undergo clonal expansion and secrete type I (IFN- α and IFN- β) and type II (IFN- γ) interferons as well as tumor necrosis factor (TNF), while activated macrophages secrete interleukin 6 (IL-6), TNF, and nitric oxide (NO). These cytokines may further contribute to apoptosis and necrosis in bystander splenocytes, culminating in IMS (8, 11) (**Figure 1**). However, the precise molecular mechanisms of THEV-induced IMS or the relevant intracellular signaling pathways are poorly understood (6). Elucidating the specific mechanisms and pathways of THEV-induced IMS is a crucial step in THEV research as it could present a means of mitigating IMS.

Bulk mRNA sequencing (RNA-seq), a next generation sequencing approach to transcriptomic studies, is a versatile, high throughput, and cost-effective technology that allows a broad survey of the entire transcriptome of a cell population, thereby uncovering the active genes and molecular pathways and processes.

68 This technology has been leveraged in an ever-increasing number of studies to elucidate active cellular
69 processes under a wide range of treatment conditions, including viral infections (12–16). In RNA-seq stud-
70 ies, differentially expressed genes (DEGs) identified by contrasting pairs of different experimental conditions
71 are key to unlocking the interesting biology or mechanism. Identified DEGs are typically used for functional
72 enrichment analyses in large curated knowledgebases such as gene ontology (GO) and Kyoto Encyclo-
73 pedia of Genes and Genomes (KEGG) pathways which connect genes to specific biological processes,
74 functions, and pathways, shedding light on the biological question under study (17, 18).

75 To our knowledge, no study has used RNA-seq to elucidate the molecular mechanisms and pathways
76 leading to THEV-induced IMS. To effectively counteract the immunosuppressive effect of the vaccine, it is
77 essential to unravel the host cell processes/pathways influenced by the virus to bring about IMS. In this
78 study, we present the first transcriptomic profile of THEV-infected cells using paired-end bulk RNA-seq in
79 a turkey B-cell line (MDTC-RP19), highlighting key host genes, cellular/molecular processes and pathways
80 affected during a THEV time course infection. We specifically focus on cellular processes related to cell
81 survivability that can elucidate THEV-induced IMS.

2. Materials and Methods

Cell culture and Infection

The Turkey B-cell line (MDTC-RP19, ATCC CRL-8135) was grown as a suspension culture in 1:1 complete Leibovitz's L-15/McCoy's 5A medium with 10% fetal bovine serum (FBS), 20% chicken serum (ChS), 5% tryptose phosphate broth (TPB), and 1% antibiotic solution (100 U/mL Penicillin and 100 μ g/mL Streptomycin), at 41°C in a humidified atmosphere with 5% CO₂. Infected cells were maintained in 1:1 serum-reduced Leibovitz's L15/McCoy's 5A media (SRLM) with 2.5% FBS, 5% ChS, 1.2% TPB, and 1% antibiotic solution. A commercially available THEV vaccine was purchased from Hygieia Biological Labs (VAS strain). The stock virus was titrated using an in-house qPCR assay with titer expressed as genome copy number (GCN)/mL, similar to Mahshoub *et al* (19). Cells were THEV-infected or mock-infected in triplicates or duplicates, respectively at a multiplicity of infection (MOI) of 100 GCN/cell, incubated at 41°C for 1 hour, and washed three times with phosphate buffered saline (PBS) to remove unattached virus particles. At each time point (4-, 12-, 24-, and 72-hpi), triplicate (THEV-infected) and duplicate (mock-infected) samples were harvested for total RNA extraction.

RNA Extraction and Sequencing

Total RNA was extracted from infected cells using the ThermoFisher RNAqueous™-4PCR Total RNA Isolation Kit (which includes a DNase I digestion step) per manufacturer's instructions. Agarose gel electrophoresis was performed to check RNA integrity (data not shown). The RNA quantity and purity was initially assessed using Nanodrop, and RNA was used only if the A260/A280 ratio was 2.0 ± 0.05 and the A260/A230 ratio was >2 and <2.2 . Extracted total RNA samples were sent to LC Sciences, Houston TX for poly-A-tailed mRNA sequencing. RNA integrity was checked with Agilent Technologies 2100 Bioanalyzer High Sensitivity DNA Chip and samples with an RNA integrity number (RIN) < 7 were excluded. Poly(A) RNA-seq library was prepared following Illumina's TruSeq-stranded-mRNA sample preparation protocol. Paired-end sequencing, generating 149 bp reads was performed on the Illumina NovaSeq 6000 sequencing system. The paired-end 149 bp sequences obtained during this study and all expression data have been submitted to the Gene Expression Omnibus database, under accession no GSE286211

Quality Control and Mapping Process

Sequencing reads were processed following a well-established protocol described by Pertea *et al* (20), using Snakemake - version 7.32.4 (21), a popular workflow management system to drive the pipeline.

Briefly, raw sequencing reads were trimmed with Cutadapt - version 1.10 (22) and the quality of trimmed reads evaluated using the FastQC software, version 0.12.1 (Bioinformatics Group at the Babraham Institute, Cambridge, United Kingdom; www.bioinformatics.babraham.ac.uk), achieving an overall Mean Sequence Quality (PHRED Score) of 36. Trimmed reads were mapped to the reference turkey (*Meleagris gallopavo*) genome file GCF_000146605.3_Turkey_5.1_genomic.fna.gz from NCBI (genome build: melGal5) (https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/146/605/GCF_000146605.3_Turkey_5.1/) with Hisat2 - version 2.2.1 (20) using the accompanying gene transfer format (GTF) annotation file (GCF_000146605.3_Turkey_5.1_genomic.gtf.gz) to build a genomic index. Samtools - version 1.21 was used to convert the output Sequence Alignment Map (SAM) file to the Binary Alignment Map (BAM) format. The StringTie (v2.2.1) software (20), set to expression estimation mode was used to generate normalized gene expression estimates from the BAM files for genes in the reference GTF file after which the prepDE.py3 script was used to extract read count information from the StringTie gene expression files, providing an expression-count matrix for downstream DEG analysis.

DEG Analysis and Functional Enrichment Analysis

DEG analysis between mock- and THEV-infected samples was performed using DESeq2 (23), which employs a Negative Binomial distribution model for determining statistical significance when comparing read counts from multiple replicates. Genes with (FDR)-adjusted P-value ≤ 0.05 were considered as differentially expressed. The sequencing data (FASTQ files), expression-count matrix, and DEG analysis results from DESeq2 are deposited at NCBI Gene Expression Omnibus under accession number GSE286211. The functional profiling of DEGs (GO and KEGG analyses) were performed based on GO databases and KEGG databases using DAVID and the R package gprofiler2 (24) with *M. gallopavo* as the reference organism. Results with P_{adjusted} -value ≤ 0.05 were included as functionally enriched. All visualization plots were made using ggplot2, pheatmap, and ggvenn R packages (25–27).

Validation of DEGs by Reverse Transcriptase Quantitative PCR (RT-qPCR)

The gene expression levels of representative DEGs (*APAF1*, *BMF*, *FADD*, *PDCD4*, *MADD*, *VCP*, *UFD1*, *EDEM1*, *EIF3D*, *EIF3M*, *RPL8*, *RPL10A*) were validated by quantification of relative mRNA levels with turkey *GAPDH* mRNA levels as the control gene. Briefly, the cells were infected and RNA extracted as described for the RNA sequencing samples with three biological replicates at 12- and 24-hpi each for both THEV-infected or mock-infected samples. First-strand cDNA synthesis of total RNA was performed with an oligo-dT primer to amplify poly-A-tailed mRNA using SuperScript™ IV First-Strand Synthesis System. The

parent RNA was digested using RNase H after cDNA synthesis was complete to ensure that only cDNA remained as the template for the RT-qPCR quantification. The RT-qPCR was performed with the PowerUp™ SYBR™ Green master mix from Applied Biosystems with primers designed manually in the SnapGene software. The primers were checked for specificity using NCBI Nucleotide BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastn>) before use. All primers used in this study are listed in **Supplementary Table S1**. Relative mRNA levels were calculated by $2^{-\Delta\Delta CT}$ method (28).

Statistical Analysis

Statistical analyses of the RT-qPCR results were performed using R (Version 4.3.3) with Student's t-test and Mann-Whitney U test for the comparison between two groups. A p-value ≤ 0.05 was considered statistically significant.

3. Results

3.1.1 Sequencing Results

To identify the host transcriptomic profile during THEV infection, we infected MDTC-RP19 cells with THEV or no virus (mock) in triplicates or duplicates, respectively, and harvested total RNA at 4-, 12-, 24-, and 72-hours post infection (hpi). mRNAs extracted from mock- or THEV-infected cells were sequenced on the Illumina platform, yielding a total of **776.1** million raw reads (149 bp in length) across all samples (see **Table 1** for sequencing statistics). After trimming low-quality reads, the remaining **742.8** million total paired-end trimmed reads (approximately, **34.7-47.9** million reads per sample) were mapped to the reference genome of *M. gallopavo* obtained from the National Center for Biotechnology Information (NCBI). The percentage of reads that mapped to the host genome across all samples ranged from **32.4** to **89.2%**. We observed that the fraction of reads that mapped to the host genome decreased while those mapping to the virus genome increased over the course of the infection as the viral infectious cycle progressed. Despite excellent quality scores at all time points (**Table 1**), DEGs identified at 4- and 72-hpi did not yield any statistically significant results in the downstream functional enrichment analyses (GO term and KEGG pathway analysis) and they were excluded from all subsequent analyses. In the remaining 12- and 24-hpi samples, a high consistency was observed between biological replicates (**Figure 2A and B**).

3.1.2 DEGs of THEV-infected Versus Mock-infected Cells

We quantified gene expression levels with the StringTie software (20) in Fragments per kilobase of transcript per million (FPKM) units. The differential expression analysis of DEGs was performed with the DESeq2 R package (23) which employs a negative binomial distribution model for determining statistical significance. Using a false discovery rate (FDR)-adjusted P-value cutoff ≤ 0.05 as the inclusion criteria, **2,343** and **3,295** genes from THEV-infected samples were identified as differentially expressed relative to their time-matched mock-infected samples at 12-hpi and 24-hpi, respectively. The results from the DEG analyses at 12- and 24-hpi have been deposited in NCBI Gene Expression Omnibus (<http://www.ncbi.nlm.nih.gov/geo>) under accession number GSE286211 with files named total_12hrsDEGs.csv.gz and total_24hrsDEGs.csv.gz, respectively. We compared THEV-infected samples relative to their time-matched mock-infected samples in identifying the significant DEGs and in the functional enrichment analyses. At 12-hpi (THEV-infected versus mock-infected), we found **1,079** upregulated genes and **1,264** downregulated genes, whereas **1,512** genes were upregulated and **1,783** genes downregulated at 24-hpi (THEV-infected versus mock-infected) (**Figure 2C**, and **Figure 3A-C**). The log₂fold-change (FC) values at 12-hpi ranged between **-1.4** and **+1.7** for **TMEM156** (Transmembrane Protein 156) and **LIPG** (Lipase G), respectively. At 24-hpi, the log₂FC values ranged between **-2.0** and **+2.6** for **C1QTNF12** (C1q And TNF Related 12) and **KCNQ1** (Potassium

Voltage-Gated Channel Modifier Subfamily G Member 1), respectively.

3.1.3 Functional Enrichment Analyses (GO and KEGG pathway Analyses)

Gene ontology (GO) enrichment analysis was performed for the DEGs determined at the 12- and 24-hpi timepoints with the DAVID (Database for Annotation, Visualization and Integrated Discovery; version 2021) online resource (29) and the gprofiler2 R package – version **0.2.3** (24), which outputs results according to the three branches of the GO directed acyclic graph – cellular components (CP), biological processes (BP), and molecular functions (MF). We compared THEV-infected samples relative to their time-matched mock-infected samples for each timepoint. Results with $P_{\text{adjusted-value}} \leq 0.05$ were considered functionally enriched. The GO enrichment analyses results at 12-hpi and 24-hpi showed significant overlaps among all three GO categories. At both time points, cellular breakdown processes were upregulated while cellular maintenance processes and structures were downregulated in all three GO categories (**Table 2A-B** and **Table 3A-B**).

For upregulated DEGs at 12-hpi, we observed that the GO terms annotated under the BP category broadly cluster into: apoptosis and autophagy, cellular metabolism (catabolic processes), sterol biosynthesis, response to stimuli, and protein processing (**Figure 4A** and **Table 2A**). In the CC category, the GO terms relate primarily with cytoplasmic vacuolation, while in the MF category, they broadly fit under protein binding and kinase activity (**Table 2A**). For downregulated DEGs at 12 hpi, the GO terms in BP category generally fell under translation, protein biosynthesis and folding, ribosome biogenesis, nitrogen compound metabolism, nucleic acid synthesis, repair, metabolism, processing, and replication, and energy metabolism. Also, immunoglobulin production and isotype switching were downregulated (**Figure 4C** and **Table 2B**). In the CC category GO terms broadly grouped into ribosome, mitochondria, respirasome, nucleus, and spliceosome, while in the MF category, they generally belong to translation regulator activity, protein folding chaperone, catalytic activity (acting on a nucleic acids), and ATP hydrolysis activity (**Table 2B**).

At 24-hpi, we found that the GO terms in the BP category for upregulated DEGs were associated with apoptosis and autophagy, lipid and sterol biosynthesis, catabolic process, protein ubiquitination and proteolysis, cell signaling, and cell metabolism. Additionally, host defense response and genes that negatively regulate cytokine production were upregulated (**Figure 4B** and **Table 3A**). In the CC category, the GO terms were related to cytoplasmic vacuolation and the lysosome, similar to those identified at 12-hpi. In the MF category, the GO terms grouped into protein ubiquitination activity, kinase and acyltransferase activity, and macromolecule binding activity (**Table 3A**). GO terms for the downregulated DEGs were markedly similar to those at 12-hpi in all three GO categories. In the BP category, the GO terms broadly group into trans-

lation, peptide biosynthesis and folding, ribosome biogenesis, aerobic respiration and ATP synthesis, and cell cycle process and nucleic acid replication and processing (**Figure 4D** and **Table 3B**). The GO terms in the CC category group under ribosome, mitochondrion, nucleus and chromosomes, while the MF category, the GO terms grouped into structural components of ribosome and translation regulator activity, catalytic activity acting on a nucleic acid and nucleic acid binding, aminoacyl-tRNA ligase activity, and NAD binding (**Table 3B**).

KEGG pathway analysis on the DEGs was also performed using both the gprofiler2 R package (24) and the DAVID online resource. Both resources gave similar results, but the results from DAVID (**Table 4**) included more information than the gprofiler2 results (**Table S2**). The results from the KEGG pathway analysis were consistent with the GO results, revealing that generally, cell maintenance and upkeep pathways were downregulated while cell death and breakdown pathways were upregulated. We observed that cell maintenance pathways such as DNA replication and repair, ribosome biogenesis, spliceosome, and oxidative phosphorylation were downregulated at both 12- and 24-hpi. Pathways such as: autophagy, response to virus (Influenza A), and steroid biosynthesis were upregulated at 12-hpi, which is similar to 24-hpi, where pathways such as: autophagy, ubiquitin-mediated proteolysis, lysosome, protein processing in endoplasmic reticulum, and steroid biosynthesis were upregulated.

It is well-established that THEV induces cell death (apoptosis and necrosis) in infected B-cells, which is linked to THEV-induced IMS (8, 11, 30). Hence, we were particularly interested in cellular processes and pathways associated with cell death and pathways that may affect the survival of the host B-cells, thereby accounting for THEV-induced IMS. We highlight the upregulated cell death (apoptosis and autophagy), ubiquitin-dependent endoplasmic reticulum [ER]-mediated protein degradation, and suppressed cell maintenance pathways as well as cytokine deregulation identified by our GO and KEGG analyses as the likely key aspects of THEV-host cell interaction relevant to THEV-induced IMS.

3.1.4 Cell Death and Breakdown Pathways Upregulated by THEV

Many virus families, including adenoviruses, herpesviruses, poxviruses, baculoviruses, parvoviruses, retroviruses, rhabdoviruses, paramyxoviruses, orthomyxoviruses, togaviruses, and picornaviruses are known to trigger apoptosis in infected host cells either through direct viral protein action or the host antiviral response (31–33). Our data show that apoptotic and autophagic pathways are upregulated during THEV infection, supporting previous findings of apoptosis and necrosis of THEV-infected cells (8, 11, 30). For example, several proapoptotic members of the BCL2 (B-cell lymphoma 2) protein family such as BCL2 antagonist/killer 1 (*BAK1*), BCL2 interacting protein 3 like (*BNIP3L*), BCL2 interacting protein 3 (*BNIP3*), and Bcl2 modifying factor (*BMF*) were upregulated. Additionally, Fas cell surface death receptor (*FAS*),

Fas associated via death domain (*FADD*), MAP kinase-activating death domain (*MADD*), programmed cell death 4 (*PDCD4*), RB1 inducible coiled-coil 1 (*RB1CC1*), activating transcription factor 4 (*ATF4*), receptor interacting serine/threonine kinase 1 (*RIPK1*), tumor necrosis factor receptor superfamily member 1B (*TNFRSF1B*), pro-apoptotic WT1 regulator (*PAWR*), and apoptotic peptidase activating factor 1 (*APAF1*), which are potent proapoptotic factors were upregulated at both timepoints. Interestingly, both the intrinsic (*BAK1*, *BNIP3L*, *BNIP3*, *BMF*, *RB1CC1*, *ATF4*, *PDCD4*, and *APAF1*) and extrinsic (*FAS*, *FADD*, *TNFRSF1B*, *MADD*, and *RIPK1*) apoptotic pathways were represented. Conversely, several anti-apoptotic proteins such as BCL2 apoptosis regulator (*BCL2*), BCL2 interacting protein 2 (*BNIP2*; interacts directly with adenovirus E1B-19K protein), BCL2 related protein A1 (*BCL2A1*), and apoptosis inhibitor 5 (*API5*) were also upregulated. Thus, apoptosis and its regulation pathways are clearly upregulated; this highlights the host-virus tug-of-war also typical in Mastadenovirus infections. Moreover, several genes associated with autophagy such as: TNF receptor associated factor 6 (*TRAF6*), autophagy related 9A (*ATG9A*), unc-51 like autophagy activating kinase 2 (*ULK2*), and autophagy related 4B cysteine peptidase (*ATG4B*) were upregulated.

3.1.5 Downregulation of Cell Maintenance Pathways

Forcibly transitioning the host cell cycle to the S phase during the early phase of infection is a prerequisite for a productive adenovirus infection (34). Interaction of the viral E1A early proteins with the host pRb (retinoblastoma) protein releases the host transcription factor E2F, which activates genes required for S phase cell cycle induction. Viral E1A also binds the host transcriptional co-activator p300/CBP (34, 35). Our GO and KEGG pathway results showed that at 12-hpi, several key genes involved with cell cycle transition were upregulated. Notably, E1A binding protein p300 (*EP300*), cyclin genes (*CCND3*, *CCNG1*, *CCNG2*, *CDK6*), anaphase promoting complex subunit 1 (*ANAPC1*), and cell division cycle 27 (*CDC27*) were upregulated. However, unlike the observation in Mastadenoviruses, the cell cycle regulation at 12- and 24-hpi seem complicated as some key cell cycle-related genes and general cell maintenance processes were concurrently downregulated.

We found that several essential cell maintenance processes whose suppression can trigger apoptosis were downregulated. Severe DNA damage is a known mechanism of apoptosis induction, called DNA damage-dependent apoptosis (36). Repression of host RNA and protein synthesis is also strongly associated with apoptosis (37). Several processes related to DNA and RNA synthesis, maintenance, and repair such as nucleotide biosynthesis and metabolism, double strand break repair, DNA excision repair, RNA biosynthesis, RNA processing, DNA replication, mitotic cell cycle process, protein-RNA complex organization, and DNA damage response were downregulated at both timepoints. Notable genes identified include DNA ligase 1 (*LIG1*), X-ray repair cross complementing 1 (*XRCC1*), cyclin dependent kinase 1 and 2 (*CDK1*, *CDK2*),

checkpoint kinase 1 (*CHEK1*), 8-oxoguanine DNA glycosylase (*OGG1*), BLM RecQ-like-helicase (*BLM*), BRCA1 DNA repair associated (*BRCA1*), and several RAD family proteins (*RAD21*, *RAD51*, *RAD51B*, *RAD51C*, *RAD54B*).

Protein synthesis-related processes, including ribosome biogenesis, rRNA processing, ribosome assembly, protein folding, translational initiation, protein maturation, ribosome and ribonucleoprotein complex formation, translation pre-initiation complex formation, and cytoplasmic translation were significantly downregulated at both 12- and 24-hpi. Notable genes identified include eukaryotic translation initiation factors (*EIF1*, *EIF1AX*, *EIF3E* and *EIF3F*, *EIF3H*, *EIF3I*, *EIF3L* and *EIF3M*), biogenesis of ribosomes BRX1 (*BRX1*), MCTS1 re-initiation and release factor (*MCTS1*), and ribosomal protein subunits (*RPL8*, *RPL10a*, *RPL11*, *RP12*, *RP13*, *RP14*, *RP15*, *RP18a*, *RP19*).

3.1.6 Endoplasmic Reticulum (ER) Stress Response during THEV infection

Our KEGG pathway analysis (**Table 4**) showed that protein processing in the ER, and ubiquitin-mediated proteolysis are significantly upregulated (**Figure 5**). The GO results (**Table 3A**) showed that specifically, ER stress and the ER-associated protein degradation (ERAD) pathway, a branch of the unfolded protein response (UPR) were upregulated during THEV infection, especially at 24-hpi. The ER is the major site for protein synthesis, folding and quality control, and sorting (38). Upon ER stress or continued accumulation of unfolded proteins in the ER lumen, the UPR pathways are activated to restore ER homeostasis. The ERAD pathway, a ubiquitin-proteasome-dependent pathway, is a protein quality control system activated for degradation of misfolded and unassembled proteins (38). In our results, the THEV-infected samples showed significant increase in ERAD pathway effector proteins, such as valosin containing protein (*VCP*), ubiquitin recognition factor in ER associated degradation 1 (*UFD1*), ER degradation enhancing alpha-mannosidase like proteins 1 and 3 (*EDEM1*, *EDEM3*), cullin 1 (*CUL1*), and ubiquitin 1 (*UBQLN1*). Other genes related to other UPR pathways such as *HSPA5* and *ATF4* were also upregulated. Our KEGG pathway (**Table S2**) and GO (**Figure 4B**) results indicated a significant upregulation of ubiquitin-mediated proteolysis with other ubiquitination pathway proteins such as ubiquitin conjugating enzymes (*UBE2J2*, *UBE2E3*, *UBE2Z*), ubiquitin protein ligases (*UBE3A*, *UBE3B*), NPL4 homolog ubiquitin recognition factor (*NPLOC4*), and ubiquitin-like modifier activating enzyme 6 (*UBA6*) showing significant upregulation. Additionally, the heat shock family of chaperone proteins such as the DnaJ heat shock protein family (*HSP40*) members (*DNAJB11*, *DNAJB12*, *DNAJB2*, *DNAJC10*), heat shock protein family A (*HSP70*) members (*HSPA4L*, *HSPA5*, *HSPA8*), and heat shock protein 90 alpha family class A member 1 (*HSP90AA1*) were upregulated at 24-hpi. Moreover, the KEGG pathway analysis (**Table 4**) shows a significant upregulation in lysosome formation, lumen acidification, and lysosomal degradation, likely an indication of ER-to-lysosome-associated degradation. Taken

together, these results suggest that THEV infection triggers significant ER-associated protein degradation, which may contribute to cell death and IMS.

3.1.7 Differential Expression of Cytokine and Cytokine Receptor-encoding Genes

Our KEGG pathway results showed that a pathway similar to immune response to influenza A infection was upregulated at 12-hpi. Our GO analysis also identified terms such as regulation of lymphocyte activation and regulation of cytokine production as upregulated at both 12- and 24-hpi. Genes involved include *IL18*, *IL2RB*, *IL4R*, *IL5RA*, TNF receptor associated factors (*TRAF2*, *TRAF3*, *TRAF6*, *TRAF7*, *TRAFD1*), TNF receptor superfamily members (*TNFRSF1B*, *TNFRSF8*, *TNFSF4*), interferon-induced with helicase C domain 1 (*IFIH1*), interferon-induced double-stranded RNA-activated protein kinase (*PKR*), and *CD80*. In contrast, cytokine inhibitors such as suppressor of cytokine signaling (*SOCS3* and *SOCS5*) were also upregulated at both 12 and 24-hpi and immunoglobulin production and isotype switching GO terms were downregulated at 12-hpi. This inconsistency is likely an indicator of the struggle between the virus and its host. While several cytokines were regulated by THEV as in the proposed model of THEV immunopathogenesis (**Figure 1**), the cytokines in the model (IFN- α , IFN- β , IFN- γ , TNF, and IL-6) were not significantly differentially expressed in our data. However, some of the differentially expressed cytokines and cytokine receptors (*TNFRSF8*, *TRAF7*) identified in this study are positive regulators of apoptosis; therefore, they may play a role in THEV-induced IMS.

3.1.8 Validation of DEGs by Reverse Transcriptase Quantitative PCR (RT-qPCR)

To validate the RNA-seq results, 12 DEGs (8 upregulated and 4 downregulated) were selected for RT-qPCR. The DEGs were representative of apoptosis (*APAF1*, *BMF*, *FADD*, *MADD*, and *PDCD4*), ERAD and ubiquitination (*VCP*, *UFD1*, *EDEM1*), and ribosome biosynthetic (*EIF3D*, *EIF3M*, *RPL8*, *RPL10A*) pathways. As shown in **Figure 6**, the RT-qPCR results corroborate the RNA-seq results, further reinforcing the validity of the RNA-seq transcriptomic profile results. Although there was no inconsistency between the RNA-seq and RT-qPCR results in terms of gene regulations, the fold changes in the RT-qPCR results were consistently higher than observed in the RNA-seq results. Our RT-qPCR primers showed excellent target specificity; only one amplicon of the expected size was amplified as shown by the melt curves (data not shown) and gel electrophoresis (**Figure S1**). According to our Student's T-test and Mann-Whitney U test, the difference in gene expression levels in all the selected genes were statistically significant.

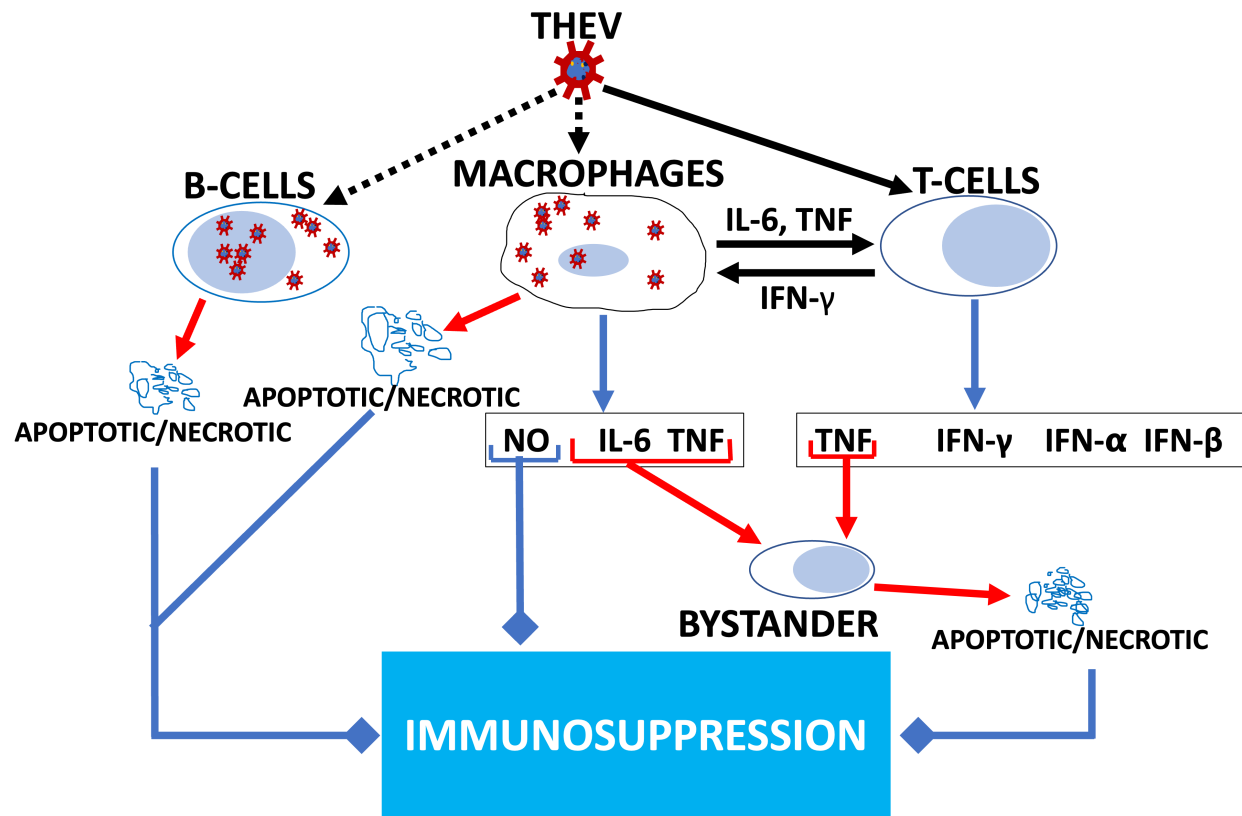


Figure 1. Model of THEV-induced immunosuppression in turkeys. THEV infection of target cells is indicated with black dotted arrows. Black unbroken arrows indicate cell activation. Red arrows indicated signals leading to cell death (apoptosis/necrosis). Blue arrows indicate all cytokines released by the cell. Blue arrows with square heads indicated an event leading to IMS. Adapted from Rautenschlein *et al.* (8).

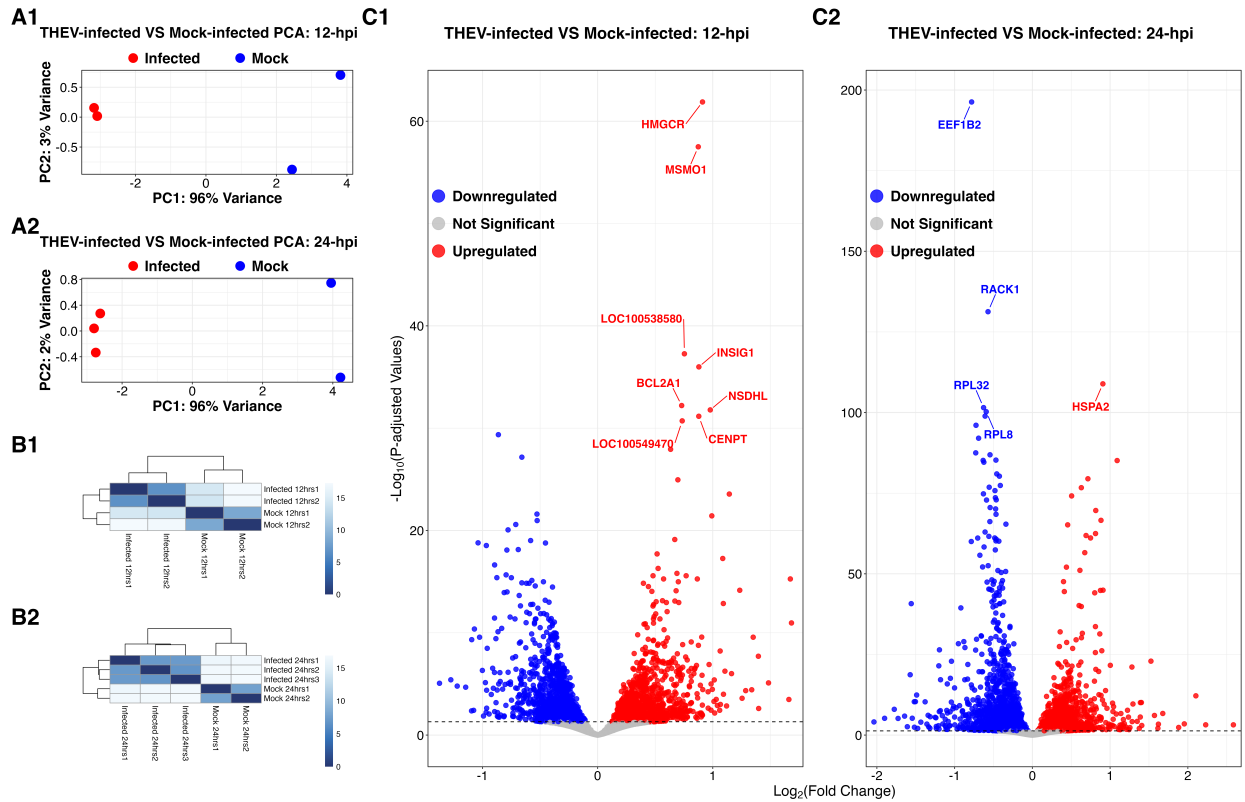


Figure 2. (A) Principal component analysis (PCA) of turkey B-cells during THEV infection. At 12-hpi (**A1**), the results indicate that the first (PC1) and second (PC2) principal components account for 96% and 3% of the variation in the samples, respectively. Whereas PC1 and PC2 account for 96% and 2% of the variation, respectively at 24-hpi (**A2**). (**B**) Poisson distance matrices illustrating the RNA-seq library integrity within treatment (infected versus mock) groups. The color scale represents the distances between biological replicates for both 12-hpi samples (**B1**) and 24-hpi samples (**B2**). Dark colors represent high correlation (similarity) between the samples involved. (**C**) Volcano plots of DEGs between THEV-infected versus mock-infected cells at 12- and 24-hpi. Red, blue, and grey dots represent upregulated, downregulated, and non-significant genes, respectively for both 12-hpi samples (**C1**) and 24-hpi samples (**C2**).

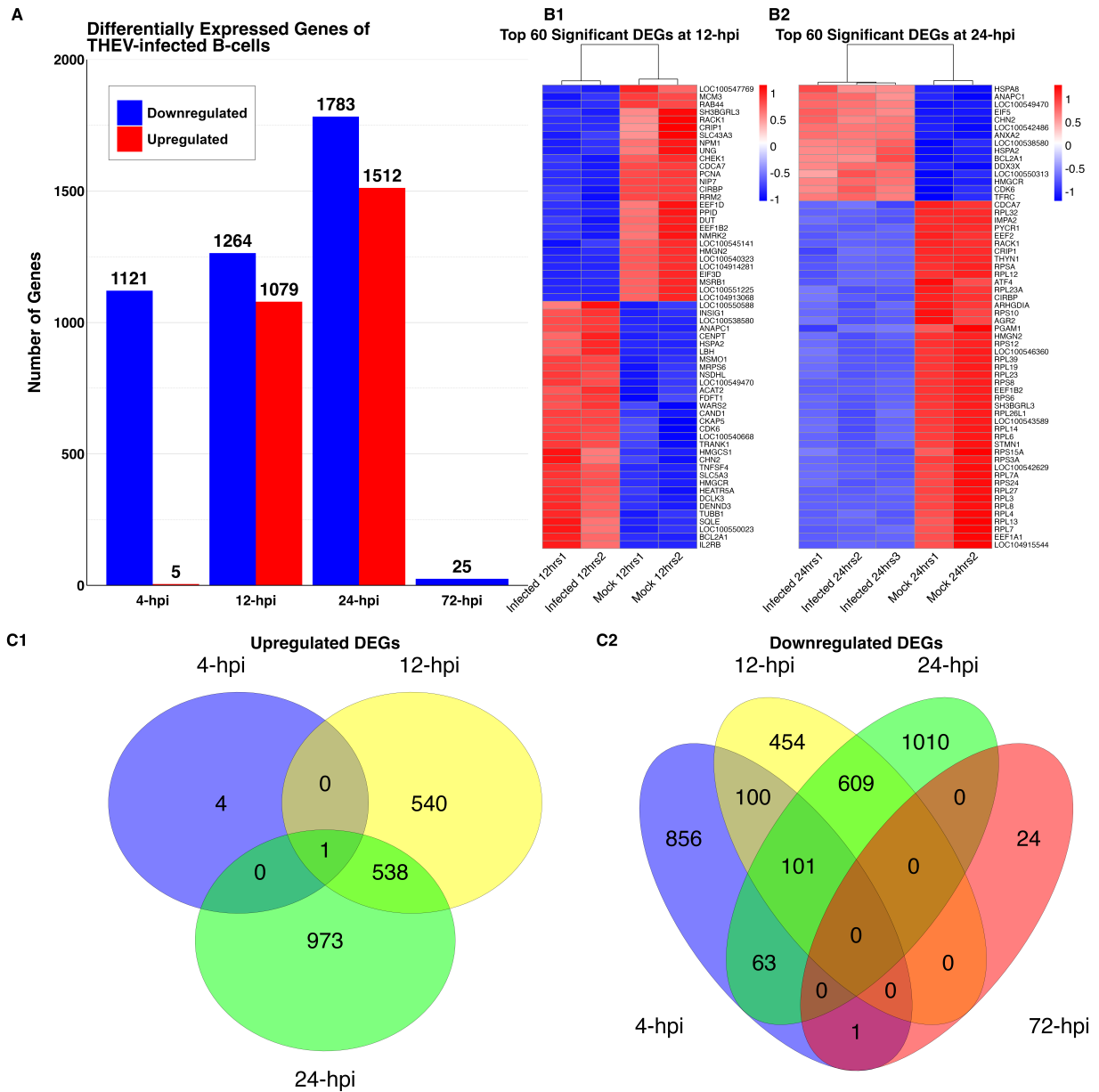


Figure 3. DEGs of THEV-infected versus mock-infected samples at different time points. **(A)** Bar plot of number DEGs identified. Red represents upregulated genes and blue represents downregulated genes. **(B)** Heatmaps of scaled expression data (Z-scores) of DEGs. DEGs identified at 12-hpi are shown in **(B1)** and DEGs at 24-hpi in **(B2)**. **(C)** Venn diagrams showing the number of DEGs identified at different time points. For the upregulated genes **(C1)**, the blue circle represents genes at 4-hpi, the yellow circle, 12-hpi, and the green circle, 24-hpi. For the downregulated genes **(C2)**, the red circle represents genes at 72-hpi, while all the other time points retain the colors from **(C1)**.

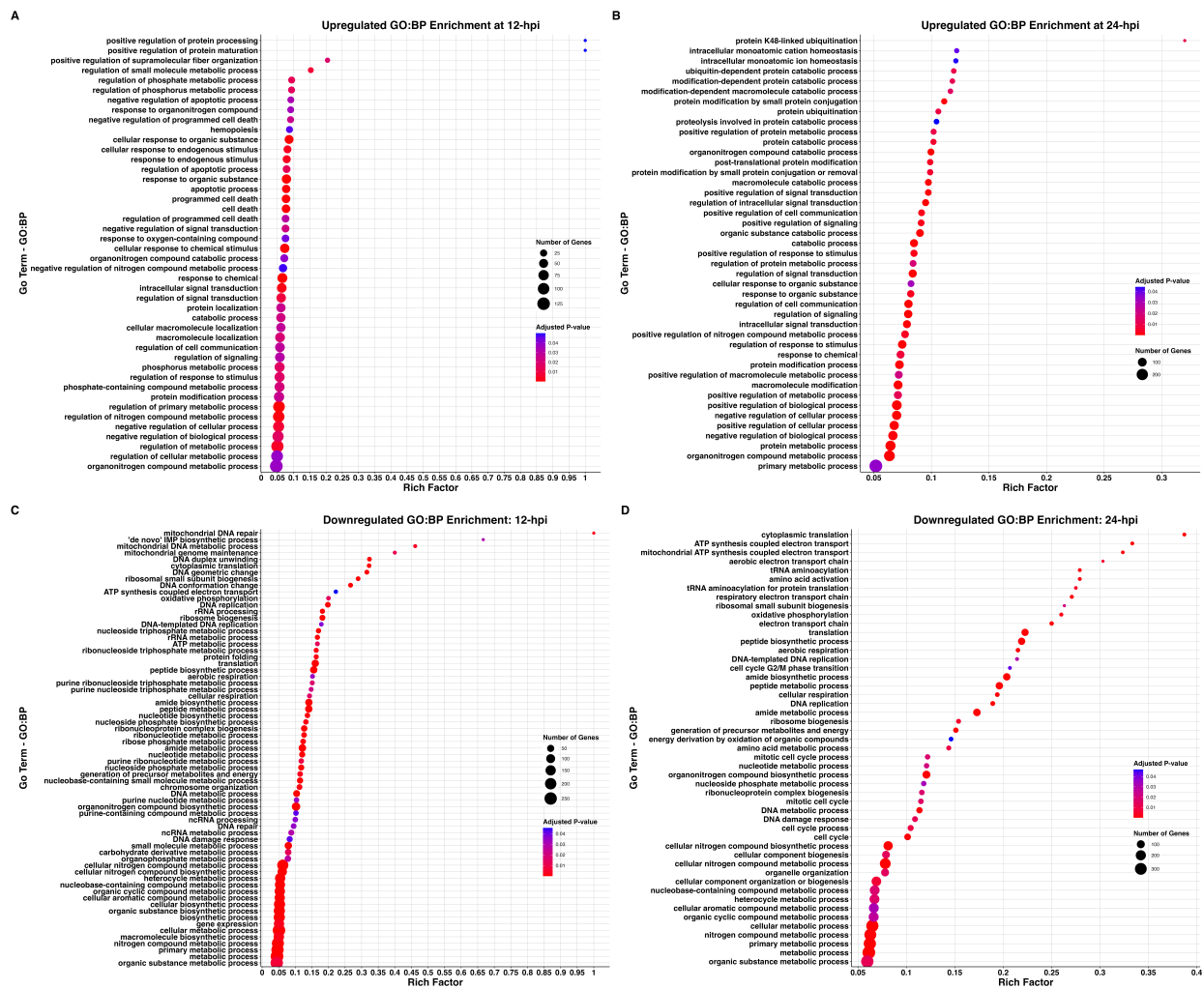


Figure 4. Dotplot of Enriched Gene Ontology Biological Processes (BP). Significant BP GO terms identified for upregulated DEGs at 12-hpi and 24-hpi are shown in **(A)** and **(B)**, respectively. Significant BP GO terms for downregulated DEGs at 12-hpi and 24-hpi are shown in **(C)** and **(D)**, respectively. The y-axis indicates GO terms and the x-axis represents the rich factor, which indicates the ratio of the number of DEGs annotated to the term to the total number of genes annotated to the term. The diameter indicates the number of genes overlapping the gene ontology term and the color indicates the enrichment P-value.

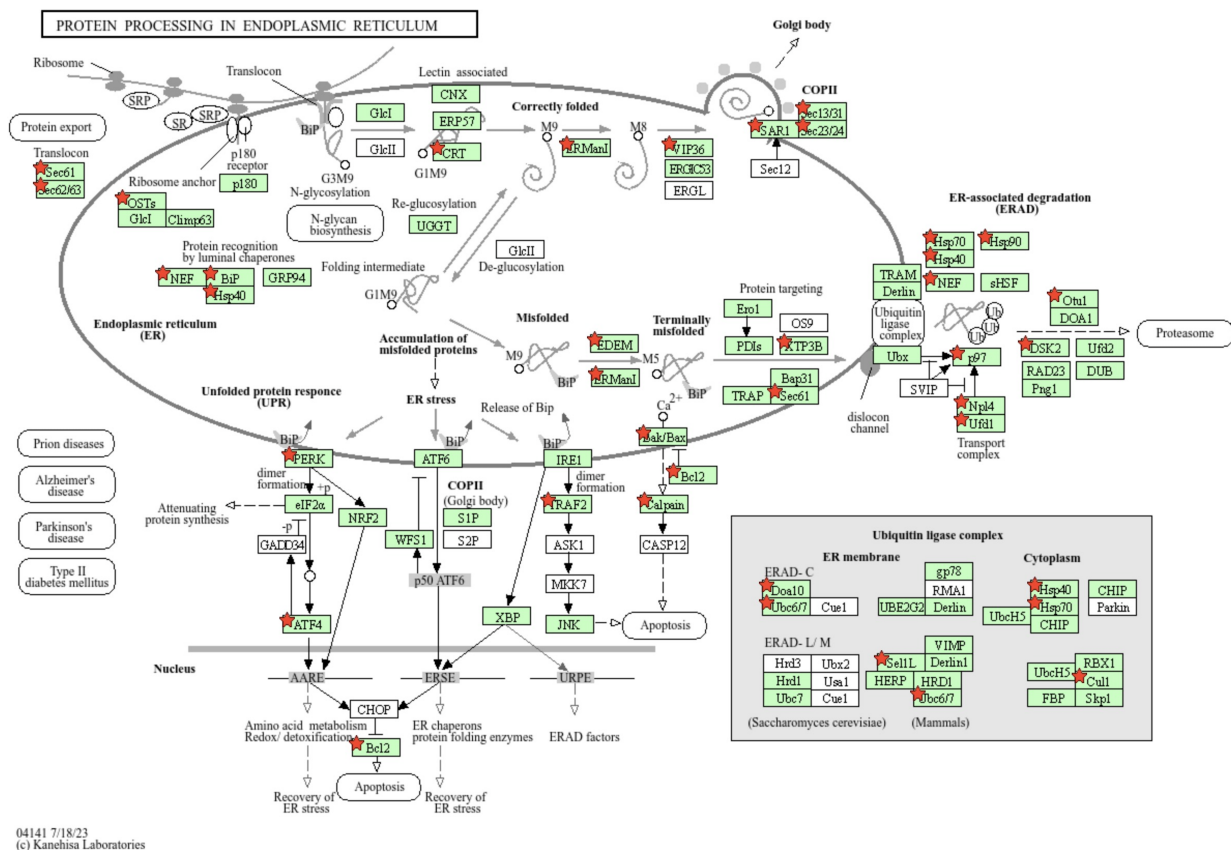


Figure 5. Upregulation of ER Unfolded Protein Response (UPR). KEGG Pathway analysis shows multiple key genes involved in the ER UPR were upregulated. All genes from our DEG list are annotated with the red star. Notably, *ATF4*, *PERK*, *VCP* (*p97*), *TRAF2*, *UFD1* and several *BCL2* and heat shock proteins are upregulated. We see that the *PERK* branch of the UPR pathway linked to apoptosis is upregulated. Another pathway linked to apoptosis via *BAX* is shown as well as the *ERAD* protein degradation pathway. Note that due to limited annotation of the host genome, a significant proportion of the DEGs were not recognized by the database; hence not shown here. Figure generated from KEGG pathway analysis in DAVID.

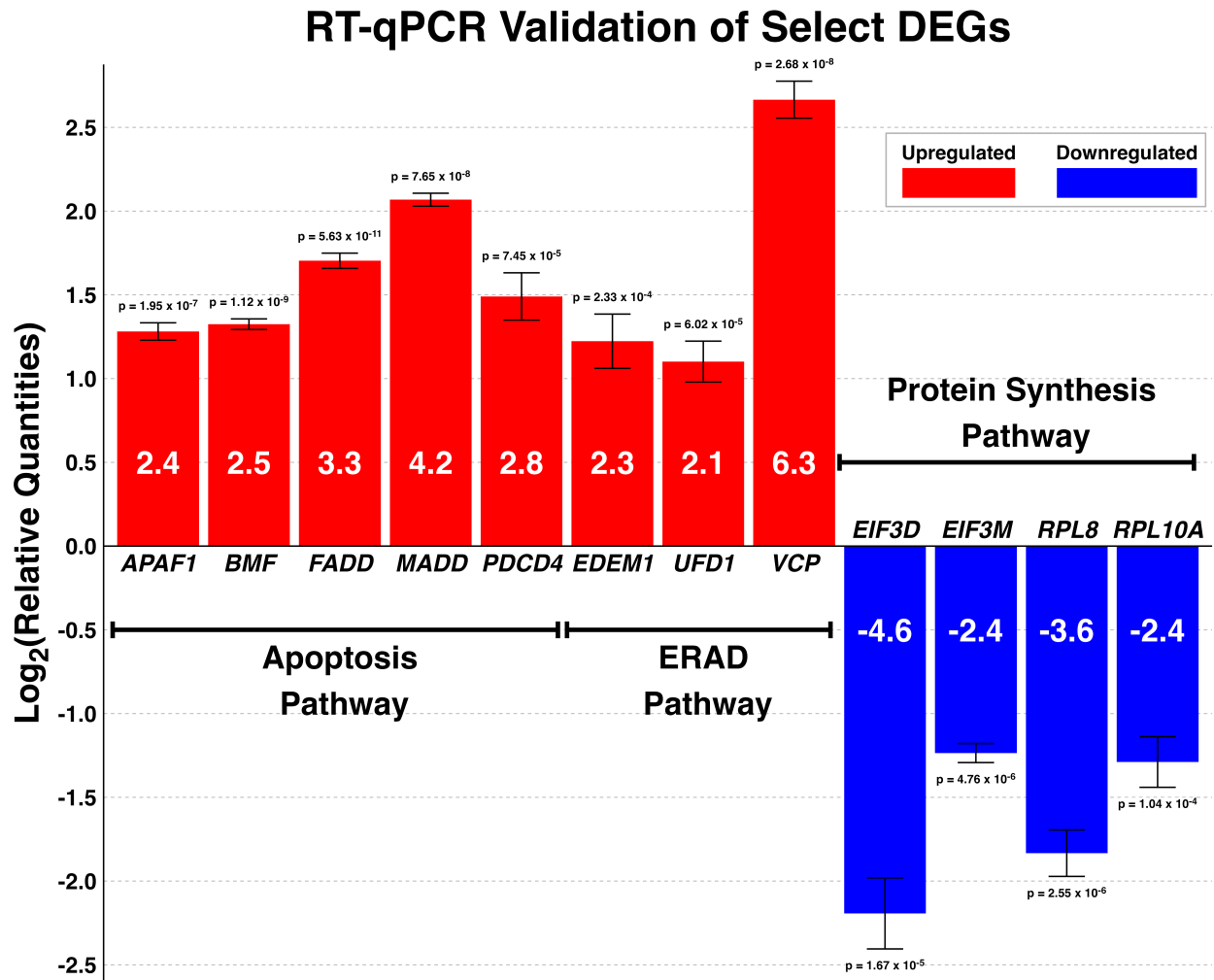


Figure 6. Validation of representative DEGs involved in Apoptosis, Protein synthesis, and ER-stress responses by RT-qPCR. MDTC-RP19 cells infected with THEV- or mock-infected were subjected to RT-qPCR analysis for the relative expression of the indicated DEGs at 24-hpi. *GAPDH* was used as the internal control. Data are expressed as the mean $\pm$ SD. All genes (THEV-infected) are statistically differentially expressed relative to their time-matched mock-infected counterparts based on Student's t-test and Mann-Whitney U test.

Table 1: Summary of sequencing, quality control, and mapping processes

Sample	Raw Reads ^M	Trimmed Reads ^M	Mapped Reads ^M	Uniquely Mapped Reads ^M	Non-uniquely Mapped Reads ^M	Q20%	Q30%	GC Content (%)
I_12hrsS1 ^{Inf}	40.6	39.0	34.7 (88.92%)	33.1 (84.78%)	1.6 (4.14%)	99.95	97.23	47.5
I_12hrsS3 ^{Inf}	38.8	37.3	33.1 (88.78%)	31.7 (84.95%)	1.4 (3.83%)	99.95	97.53	47.5
I_24hrsS1 ^{Inf}	42.7	41.0	36.2 (88.13%)	34.5 (84.2%)	1.6 (3.93%)	99.95	96.95	46.5
I_24hrsS2 ^{Inf}	42.0	40.4	35.6 (88.1%)	33.9 (83.83%)	1.7 (4.27%)	99.94	97.05	46.5
I_24hrsS3 ^{Inf}	40.5	38.9	34.2 (88.01%)	32.7 (84.12%)	1.5 (3.89%)	99.95	97.08	47.0
I_4hrsS1 ^{Inf}	39.1	37.4	33 (88.16%)	31.2 (83.43%)	1.8 (4.73%)	99.93	97.04	48.5
I_4hrsS2 ^{Inf}	41.3	39.6	35.3 (89.24%)	33.6 (84.92%)	1.7 (4.33%)	99.95	97.15	47.0
I_4hrsS3 ^{Inf}	41.5	39.8	35.5 (89.2%)	33.2 (83.29%)	2.4 (5.91%)	99.95	97.11	47.5
I_72hrsS1 ^{Inf}	41.2	39.8	28.3 (71.09%)	26.9 (67.7%)	1.3 (3.38%)	99.96	97.23	44.5
I_72hrsS2 ^{Inf}	39.3	38.0	27 (71.11%)	25.8 (67.86%)	1.2 (3.25%)	99.96	97.34	44.5
I_72hrsS3 ^{Inf}	39.9	37.1	28.3 (76.36%)	26.1 (70.3%)	2.2 (6.05%)	99.87	96.14	52.5
U_12hrsN1 ^{Mk}	42.1	40.4	35.9 (88.72%)	34.1 (84.39%)	1.7 (4.33%)	99.95	97.04	47.5
U_12hrsN2 ^{Mk}	41.0	39.3	34.7 (88.4%)	33.2 (84.53%)	1.5 (3.86%)	99.94	97.08	47.5
U_24hrsN1 ^{Mk}	38.4	37.0	32.7 (88.46%)	31.4 (84.74%)	1.4 (3.72%)	99.96	97.48	47.5
U_24hrsN2 ^{Mk}	39.9	38.4	34 (88.58%)	32.6 (84.96%)	1.4 (3.61%)	99.95	96.95	47.0
U_4hrsN1 ^{Mk}	39.4	37.9	33.7 (88.9%)	32 (84.41%)	1.7 (4.49%)	99.96	97.36	47.0
U_4hrsN2 ^{Mk}	37.6	34.7	22 (63.43%)	18.5 (53.18%)	3.6 (10.25%)	99.80	94.96	61.0
U_72hrsN1 ^{Mk}	50.3	47.9	15.5 (32.4%)	11.7 (24.5%)	3.8 (7.9%)	99.88	96.54	56.0

Sample	Raw Reads^M	Trimmed Reads^M	Mapped Reads^M	Uniquely Mapped Reads^M	Non-uniquely Mapped Reads^M	Q20%	Q30%	GC Content (%)
U_72hrsN2 ^{Mk}	40.5	38.9	34.5 (88.82%)	32.7 (84.14%)	1.8 (4.68%)	99.95	97.04	46.5

^MAll values for number of reads are in millions; ^{Inf}These are infected samples indicated by the letter 'I' and 'S' in sample names; ^{Mk}These are mock-infected samples indicated by the letters 'U' and 'N' in sample names;

Table 2A: Gene ontology analysis of Significantly Upregulated DEGs identified at 12-hpi

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
Biological Process				
GO:BP	DNA-templated transcription	2.17	26	1.53×10^{-2}
GO:BP	alcohol biosynthetic process	3.77	19	3.45×10^{-4}
GO:BP	androgen receptor signaling pathway	10.03	5	3.18×10^{-2}
GO:BP	apoptotic process	2.75	47	6.09×10^{-7}
GO:BP	apoptotic signaling pathway	3.32	20	8.19×10^{-4}
GO:BP	appendage development	4.20	9	3.40×10^{-2}
GO:BP	appendage morphogenesis	4.59	8	4.22×10^{-2}
GO:BP	autophagy	2.59	23	4.43×10^{-3}
GO:BP	biological process involved in interspecies interaction between organisms	1.80	40	1.74×10^{-2}
GO:BP	biological regulation	1.14	517	8.20×10^{-4}
GO:BP	catabolic process	1.51	108	1.03×10^{-3}
GO:BP	cell cycle	1.68	72	1.34×10^{-3}
GO:BP	cell cycle phase transition	3.29	11	4.63×10^{-2}
GO:BP	cell cycle process	1.70	59	4.85×10^{-3}
GO:BP	cell death	2.85	51	4.99×10^{-8}
GO:BP	cell division	2.20	26	1.31×10^{-2}
GO:BP	cellular catabolic process	1.64	44	4.22×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	cellular component disassembly	2.46	21	1.31×10^{-2}
GO:BP	cellular lipid biosynthetic process	9.36	7	3.16×10^{-3}
GO:BP	cellular lipid metabolic process	1.67	67	3.03×10^{-3}
GO:BP	cellular localization	1.45	145	4.34×10^{-4}
GO:BP	cellular macromolecule localization	1.58	104	4.21×10^{-4}
GO:BP	cellular metabolic process	1.23	307	8.52×10^{-4}
GO:BP	cellular response to biotic stimulus	3.21	12	3.47×10^{-2}
GO:BP	cellular response to chemical stimulus	1.56	60	2.49×10^{-2}
GO:BP	cellular response to decreased oxygen levels	4.86	8	3.18×10^{-2}
GO:BP	cellular response to hypoxia	5.02	8	2.79×10^{-2}
GO:BP	cellular response to lipid	2.66	20	8.44×10^{-3}
GO:BP	cellular response to lipopolysaccharide	3.56	11	2.92×10^{-2}
GO:BP	cellular response to molecule of bacterial origin	3.34	11	4.22×10^{-2}
GO:BP	cellular response to oxygen levels	5.02	9	1.27×10^{-2}
GO:BP	cellular response to oxygen-containing compound	1.92	33	1.98×10^{-2}
GO:BP	cellular response to stress	1.77	81	1.11×10^{-4}
GO:BP	cholesterol biosynthetic process	6.92	10	7.48×10^{-4}
GO:BP	cholesterol metabolic process	3.76	12	1.20×10^{-2}
GO:BP	deadenylation-independent decapping of nuclear-transcribed mRNA	14.33	5	8.47×10^{-3}
GO:BP	developmental growth	2.58	19	1.53×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	embryo development	1.93	29	3.43×10^{-2}
GO:BP	embryonic morphogenesis	2.36	24	9.41×10^{-3}
GO:BP	endoderm development	6.42	8	7.66×10^{-3}
GO:BP	ergosterol biosynthetic process	12.77	7	5.94×10^{-4}
GO:BP	ergosterol metabolic process	12.77	7	5.94×10^{-4}
GO:BP	establishment of localization	1.24	195	2.46×10^{-2}
GO:BP	establishment of localization in cell	1.56	104	5.94×10^{-4}
GO:BP	establishment of protein localization	1.61	73	3.79×10^{-3}
GO:BP	establishment of protein localization to organelle	2.04	34	7.41×10^{-3}
GO:BP	establishment or maintenance of cell polarity	2.51	20	1.48×10^{-2}
GO:BP	extrinsic apoptotic signaling pathway	4.18	10	1.89×10^{-2}
GO:BP	gland development	3.06	16	9.41×10^{-3}
GO:BP	growth	2.58	19	1.53×10^{-2}
GO:BP	hemopoiesis	2.16	26	1.69×10^{-2}
GO:BP	homeostasis of number of cells	3.27	14	1.27×10^{-2}
GO:BP	intracellular lipid transport	5.02	8	2.79×10^{-2}
GO:BP	intracellular protein transport	2.00	50	5.94×10^{-4}
GO:BP	intracellular signal transduction	1.54	97	1.48×10^{-3}
GO:BP	intracellular transport	1.51	79	1.02×10^{-2}
GO:BP	intrinsic apoptotic signaling pathway	3.70	12	1.31×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	limb development	4.20	9	3.40×10^{-2}
GO:BP	limb morphogenesis	4.59	8	4.22×10^{-2}
GO:BP	lipid biosynthetic process	1.94	46	1.94×10^{-3}
GO:BP	lipid metabolic process	1.53	79	7.41×10^{-3}
GO:BP	localization	1.24	219	1.25×10^{-2}
GO:BP	mRNA transcription	7.80	7	7.79×10^{-3}
GO:BP	macroautophagy	2.98	18	5.47×10^{-3}
GO:BP	macromolecule catabolic process	1.76	60	1.77×10^{-3}
GO:BP	macromolecule localization	1.58	128	3.56×10^{-5}
GO:BP	macromolecule metabolic process	1.21	286	6.45×10^{-3}
GO:BP	macromolecule modification	1.43	138	9.09×10^{-4}
GO:BP	metabolic process	1.19	426	3.93×10^{-4}
GO:BP	mitotic cell cycle	1.94	47	1.70×10^{-3}
GO:BP	mitotic cell cycle phase transition	3.34	11	4.22×10^{-2}
GO:BP	mitotic cell cycle process	2.14	41	8.20×10^{-4}
GO:BP	motor neuron apoptotic process	10.03	5	3.18×10^{-2}
GO:BP	multicellular organismal-level homeostasis	2.48	22	9.41×10^{-3}
GO:BP	negative regulation of apoptotic process	2.37	36	4.54×10^{-4}
GO:BP	negative regulation of biological process	1.56	187	1.90×10^{-7}
GO:BP	negative regulation of biosynthetic process	1.73	68	1.03×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	negative regulation of cellular biosynthetic process	1.74	68	8.52×10^{-4}
GO:BP	negative regulation of cellular metabolic process	1.79	80	9.46×10^{-5}
GO:BP	negative regulation of cellular process	1.59	174	1.90×10^{-7}
GO:BP	negative regulation of gene expression	2.16	40	8.52×10^{-4}
GO:BP	negative regulation of intracellular signal transduction	2.07	28	1.78×10^{-2}
GO:BP	negative regulation of macromolecule biosynthetic process	1.70	65	2.25×10^{-3}
GO:BP	negative regulation of macromolecule metabolic process	1.66	82	7.20×10^{-4}
GO:BP	negative regulation of metabolic process	1.70	91	1.10×10^{-4}
GO:BP	negative regulation of programmed cell death	2.35	37	4.29×10^{-4}
GO:BP	nitrogen compound transport	1.57	79	4.15×10^{-3}
GO:BP	nuclear transport	2.24	22	2.79×10^{-2}
GO:BP	nuclear-transcribed mRNA catabolic process, deadenylation-independent decay	14.33	5	8.47×10^{-3}
GO:BP	nucleobase-containing compound catabolic process	2.05	24	4.40×10^{-2}
GO:BP	nucleocytoplasmic transport	2.24	22	2.79×10^{-2}
GO:BP	nucleoside bisphosphate metabolic process	3.03	13	3.39×10^{-2}
GO:BP	organic hydroxy compound biosynthetic process	2.93	20	3.11×10^{-3}
GO:BP	organonitrogen compound metabolic process	1.29	260	3.93×10^{-4}
GO:BP	organophosphate metabolic process	1.59	65	1.09×10^{-2}
GO:BP	phosphate-containing compound metabolic process	1.63	145	1.11×10^{-6}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	phosphorus metabolic process	1.63	146	1.11×10^{-6}
GO:BP	phosphorylation	1.80	74	1.96×10^{-4}
GO:BP	phytosteroid biosynthetic process	12.04	9	3.75×10^{-5}
GO:BP	phytosteroid metabolic process	12.04	9	3.75×10^{-5}
GO:BP	positive regulation of apoptotic process	2.85	24	9.34×10^{-4}
GO:BP	positive regulation of biological process	1.33	193	1.29×10^{-3}
GO:BP	positive regulation of catabolic process	2.30	27	6.45×10^{-3}
GO:BP	positive regulation of cell communication	1.55	55	4.22×10^{-2}
GO:BP	positive regulation of cellular biosynthetic process	1.44	72	4.95×10^{-2}
GO:BP	positive regulation of cellular metabolic process	1.56	98	8.52×10^{-4}
GO:BP	positive regulation of cellular process	1.34	173	1.83×10^{-3}
GO:BP	positive regulation of macromolecule metabolic process	1.46	100	5.97×10^{-3}
GO:BP	positive regulation of metabolic process	1.53	116	4.34×10^{-4}
GO:BP	positive regulation of programmed cell death	2.74	24	1.58×10^{-3}
GO:BP	positive regulation of signal transduction	1.62	51	2.85×10^{-2}
GO:BP	positive regulation of signaling	1.55	55	4.22×10^{-2}
GO:BP	primary metabolic process	1.22	380	1.08×10^{-4}
GO:BP	process utilizing autophagic mechanism	2.59	23	4.43×10^{-3}
GO:BP	programmed cell death	2.85	51	4.99×10^{-8}
GO:BP	protein catabolic process	1.66	42	4.40×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	protein localization	1.58	104	4.21×10^{-4}
GO:BP	protein localization to organelle	1.90	52	1.15×10^{-3}
GO:BP	protein metabolic process	1.27	198	8.44×10^{-3}
GO:BP	protein modification process	1.55	138	3.56×10^{-5}
GO:BP	protein phosphorylation	2.33	61	9.07×10^{-7}
GO:BP	protein transport	1.58	62	1.45×10^{-2}
GO:BP	purine nucleoside bisphosphate metabolic process	3.03	13	3.39×10^{-2}
GO:BP	regulation of DNA-templated transcription	1.35	142	7.41×10^{-3}
GO:BP	regulation of RNA biosynthetic process	1.35	142	7.59×10^{-3}
GO:BP	regulation of RNA metabolic process	1.34	154	5.97×10^{-3}
GO:BP	regulation of anatomical structure morphogenesis	1.81	32	4.74×10^{-2}
GO:BP	regulation of apoptotic process	2.15	57	2.79×10^{-5}
GO:BP	regulation of autophagy	2.61	20	1.02×10^{-2}
GO:BP	regulation of biological process	1.15	497	5.83×10^{-4}
GO:BP	regulation of biosynthetic process	1.40	210	3.56×10^{-5}
GO:BP	regulation of catabolic process	2.01	47	8.20×10^{-4}
GO:BP	regulation of cell communication	1.36	120	1.31×10^{-2}
GO:BP	regulation of cell cycle	1.67	43	3.66×10^{-2}
GO:BP	regulation of cell cycle process	1.79	33	4.65×10^{-2}
GO:BP	regulation of cellular biosynthetic process	1.41	210	2.24×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	regulation of cellular catabolic process	2.35	21	2.18×10^{-2}
GO:BP	regulation of cellular metabolic process	1.48	252	2.56×10^{-8}
GO:BP	regulation of cellular process	1.16	473	4.21×10^{-4}
GO:BP	regulation of cytokine production	2.21	23	2.61×10^{-2}
GO:BP	regulation of developmental process	1.59	73	4.88×10^{-3}
GO:BP	regulation of epithelial cell apoptotic process	5.02	8	2.79×10^{-2}
GO:BP	regulation of gene expression	1.38	201	1.06×10^{-4}
GO:BP	regulation of intracellular signal transduction	1.60	73	4.75×10^{-3}
GO:BP	regulation of leukocyte differentiation	3.00	13	3.64×10^{-2}
GO:BP	regulation of macromolecule biosynthetic process	1.39	205	6.25×10^{-5}
GO:BP	regulation of macromolecule metabolic process	1.42	248	9.05×10^{-7}
GO:BP	regulation of metabolic process	1.47	279	6.08×10^{-9}
GO:BP	regulation of mitotic cell cycle phase transition	2.34	18	4.74×10^{-2}
GO:BP	regulation of nucleobase-containing compound metabolic process	1.36	167	1.29×10^{-3}
GO:BP	regulation of phosphate metabolic process	1.79	37	2.89×10^{-2}
GO:BP	regulation of phosphorus metabolic process	1.79	37	2.89×10^{-2}
GO:BP	regulation of primary metabolic process	1.40	226	1.02×10^{-5}
GO:BP	regulation of programmed cell death	2.07	57	6.77×10^{-5}
GO:BP	regulation of protein metabolic process	1.50	59	4.97×10^{-2}
GO:BP	regulation of response to stimulus	1.39	137	3.49×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	regulation of response to stress	1.91	46	2.54×10^{-3}
GO:BP	regulation of signal transduction	1.44	110	4.85×10^{-3}
GO:BP	regulation of signaling	1.37	121	1.25×10^{-2}
GO:BP	regulation of transcription by RNA polymerase II	1.41	111	7.98×10^{-3}
GO:BP	response to chemical	1.56	95	1.24×10^{-3}
GO:BP	response to lipid	2.41	25	5.91×10^{-3}
GO:BP	response to nitrogen compound	1.97	30	2.46×10^{-2}
GO:BP	response to organonitrogen compound	2.19	29	7.59×10^{-3}
GO:BP	response to oxygen-containing compound	1.90	44	4.16×10^{-3}
GO:BP	response to stress	1.44	112	3.79×10^{-3}
GO:BP	ribonucleoside bisphosphate metabolic process	3.03	13	3.39×10^{-2}
GO:BP	secondary alcohol biosynthetic process	7.67	13	1.58×10^{-5}
GO:BP	secondary alcohol metabolic process	3.70	14	4.85×10^{-3}
GO:BP	small molecule biosynthetic process	2.38	45	3.75×10^{-5}
GO:BP	small molecule metabolic process	1.44	93	1.24×10^{-2}
GO:BP	steroid biosynthetic process	3.96	16	8.52×10^{-4}
GO:BP	steroid metabolic process	2.54	18	2.46×10^{-2}
GO:BP	sterol biosynthetic process	7.24	13	2.79×10^{-5}
GO:BP	sterol metabolic process	3.86	15	1.83×10^{-3}
GO:BP	tissue development	1.58	51	4.22×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	transport	1.24	183	3.18×10^{-2}
GO:BP	vesicle-mediated transport	1.51	80	9.41×10^{-3}
Cellular Component				
GO:CC	Golgi apparatus	1.52	69	1.40×10^{-2}
GO:CC	bounding membrane of organelle	1.70	92	3.49×10^{-5}
GO:CC	chromatin	1.84	40	9.58×10^{-3}
GO:CC	chromosome	1.58	63	1.10×10^{-2}
GO:CC	cytoplasm	1.28	590	7.06×10^{-17}
GO:CC	cytoplasmic vesicle	1.57	88	1.12×10^{-3}
GO:CC	cytoplasmic vesicle membrane	1.69	41	2.85×10^{-2}
GO:CC	cytosol	1.69	166	6.96×10^{-10}
GO:CC	early endosome	2.11	22	3.70×10^{-2}
GO:CC	endomembrane system	1.48	200	3.53×10^{-7}
GO:CC	endoplasmic reticulum	1.54	86	2.07×10^{-3}
GO:CC	endosome	1.69	48	1.37×10^{-2}
GO:CC	endosome membrane	2.02	25	3.21×10^{-2}
GO:CC	intracellular anatomical structure	1.19	774	8.23×10^{-20}
GO:CC	intracellular membrane-bounded organelle	1.29	578	1.08×10^{-16}
GO:CC	intracellular organelle	1.23	655	7.52×10^{-16}
GO:CC	intracellular organelle lumen	1.51	135	4.91×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	intracellular vesicle	1.55	88	1.62×10^{-3}
GO:CC	membrane-bounded organelle	1.26	595	7.52×10^{-16}
GO:CC	membrane-enclosed lumen	1.51	135	4.91×10^{-5}
GO:CC	nuclear lumen	1.55	119	6.06×10^{-5}
GO:CC	nucleoplasm	1.75	104	1.57×10^{-6}
GO:CC	nucleus	1.42	371	1.33×10^{-13}
GO:CC	organelle	1.21	666	2.49×10^{-14}
GO:CC	organelle lumen	1.51	135	4.91×10^{-5}
GO:CC	organelle membrane	1.59	154	3.53×10^{-7}
GO:CC	organelle subcompartment	1.56	65	1.21×10^{-2}
GO:CC	perinuclear region of cytoplasm	2.51	25	2.11×10^{-3}
GO:CC	phagophore assembly site	4.73	8	2.83×10^{-2}
GO:CC	protein-DNA complex	1.76	42	1.40×10^{-2}
GO:CC	spindle	2.04	25	2.85×10^{-2}
GO:CC	transcription regulator complex	1.99	30	1.40×10^{-2}
GO:CC	vacuole	1.82	34	2.73×10^{-2}
GO:CC	vesicle	1.51	95	1.94×10^{-3}
GO:CC	vesicle membrane	1.74	43	1.40×10^{-2}
Molecular Function				
GO:MF	ATP binding	1.33	128	2.71×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	DNA-binding transcription factor binding	2.73	22	5.83×10^{-3}
GO:MF	R-SMAD binding	10.31	5	4.03×10^{-2}
GO:MF	RNA polymerase II-specific DNA-binding transcription factor binding	2.94	18	1.05×10^{-2}
GO:MF	adenyl nucleotide binding	1.30	133	4.38×10^{-2}
GO:MF	adenyl ribonucleotide binding	1.33	130	2.71×10^{-2}
GO:MF	binding	1.09	714	4.02×10^{-4}
GO:MF	enzyme binding	2.20	102	8.60×10^{-11}
GO:MF	enzyme regulator activity	1.50	82	1.38×10^{-2}
GO:MF	identical protein binding	2.07	54	1.96×10^{-4}
GO:MF	ion binding	1.19	325	1.05×10^{-2}
GO:MF	kinase activity	1.58	86	4.95×10^{-3}
GO:MF	kinase binding	2.14	35	5.83×10^{-3}
GO:MF	manganese ion binding	5.62	9	1.05×10^{-2}
GO:MF	molecular adaptor activity	1.90	73	5.57×10^{-5}
GO:MF	myosin phosphatase activity	4.64	9	2.71×10^{-2}
GO:MF	nuclear androgen receptor binding	12.88	5	1.83×10^{-2}
GO:MF	phosphotransferase activity, alcohol group as acceptor	1.59	80	5.83×10^{-3}
GO:MF	protein binding	1.27	427	8.52×10^{-8}
GO:MF	protein domain specific binding	2.51	25	5.83×10^{-3}
GO:MF	protein homodimerization activity	2.29	23	2.51×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	protein kinase activity	1.59	67	1.20×10^{-2}
GO:MF	protein kinase binding	2.10	31	1.18×10^{-2}
GO:MF	protein serine/threonine kinase activity	1.68	44	4.27×10^{-2}
GO:MF	protein-macromolecule adaptor activity	1.98	67	5.57×10^{-5}
GO:MF	purine ribonucleoside triphosphate binding	1.28	148	4.38×10^{-2}
GO:MF	signaling adaptor activity	3.44	12	3.01×10^{-2}
GO:MF	small molecule binding	1.19	338	5.83×10^{-3}
GO:MF	transcription coregulator activity	1.93	38	1.18×10^{-2}
GO:MF	transcription factor binding	2.38	27	6.59×10^{-3}
GO:MF	transferase activity	1.30	176	1.05×10^{-2}
GO:MF	transferase activity, transferring phosphorus-containing groups	1.51	96	5.83×10^{-3}

Table 2B: Gene ontology analysis of Significantly Downregulated DEGs identified at 12-hpi

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
Biological Process				
GO:BP	'de novo' AMP biosynthetic process	10.16	5	2.26×10^{-2}
GO:BP	'de novo' IMP biosynthetic process	11.43	5	1.51×10^{-2}
GO:BP	'de novo' XMP biosynthetic process	18.29	4	1.71×10^{-2}
GO:BP	'de novo' post-translational protein folding	8.31	10	6.05×10^{-5}
GO:BP	'de novo' protein folding	7.95	10	8.96×10^{-5}
GO:BP	AMP biosynthetic process	7.84	6	1.71×10^{-2}
GO:BP	ATP biosynthetic process	6.58	9	1.26×10^{-3}
GO:BP	ATP metabolic process	4.29	19	1.86×10^{-5}
GO:BP	ATP synthesis coupled electron transport	6.10	18	1.95×10^{-7}
GO:BP	DNA damage response	2.35	69	7.26×10^{-9}
GO:BP	DNA geometric change	4.33	9	2.22×10^{-2}
GO:BP	DNA integrity checkpoint signaling	3.75	16	6.72×10^{-4}
GO:BP	DNA metabolic process	2.74	82	3.18×10^{-14}
GO:BP	DNA recombination	2.07	23	3.64×10^{-2}
GO:BP	DNA repair	2.39	51	1.44×10^{-6}
GO:BP	DNA replication	4.95	39	2.28×10^{-14}
GO:BP	DNA replication checkpoint signaling	6.36	8	5.00×10^{-3}
GO:BP	DNA replication initiation	5.23	8	1.71×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	DNA strand elongation	6.97	8	2.67×10^{-3}
GO:BP	DNA strand elongation involved in DNA replication	7.32	8	1.93×10^{-3}
GO:BP	DNA-templated DNA replication	5.29	33	8.34×10^{-13}
GO:BP	DNA-templated DNA replication maintenance of fidelity	4.43	8	3.69×10^{-2}
GO:BP	GMP biosynthetic process	9.15	6	8.32×10^{-3}
GO:BP	GMP metabolic process	7.84	6	1.71×10^{-2}
GO:BP	NADH dehydrogenase complex assembly	4.57	8	3.13×10^{-2}
GO:BP	RNA biosynthetic process	2.55	135	4.64×10^{-22}
GO:BP	RNA export from nucleus	3.77	13	4.16×10^{-3}
GO:BP	RNA localization	3.69	22	2.39×10^{-5}
GO:BP	RNA metabolic process	2.49	152	6.25×10^{-24}
GO:BP	RNA modification	2.30	19	3.26×10^{-2}
GO:BP	RNA processing	3.04	120	6.65×10^{-26}
GO:BP	RNA splicing	2.51	33	1.35×10^{-4}
GO:BP	RNA splicing, via transesterification reactions	2.70	28	2.22×10^{-4}
GO:BP	RNA splicing, via transesterification reactions with bulged adenosine as nucleophile	2.70	28	2.22×10^{-4}
GO:BP	RNA transport	3.70	18	2.32×10^{-4}
GO:BP	XMP biosynthetic process	18.29	4	1.71×10^{-2}
GO:BP	XMP metabolic process	18.29	4	1.71×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	aerobic electron transport chain	5.97	16	2.15×10^{-6}
GO:BP	aerobic respiration	5.10	29	1.17×10^{-10}
GO:BP	amino acid activation	3.25	11	3.83×10^{-2}
GO:BP	biosynthetic process	2.25	377	3.04×10^{-57}
GO:BP	carbohydrate derivative biosynthetic process	1.64	49	2.14×10^{-2}
GO:BP	carbohydrate derivative metabolic process	1.49	69	2.28×10^{-2}
GO:BP	cell cycle	1.47	69	3.10×10^{-2}
GO:BP	cell cycle DNA replication	6.33	9	1.67×10^{-3}
GO:BP	cell cycle checkpoint signaling	3.59	20	1.16×10^{-4}
GO:BP	cell cycle process	1.50	57	4.88×10^{-2}
GO:BP	cellular biosynthetic process	2.29	329	4.05×10^{-50}
GO:BP	cellular component assembly	1.34	117	2.26×10^{-2}
GO:BP	cellular component biogenesis	1.83	182	5.57×10^{-14}
GO:BP	cellular component organization or biogenesis	1.26	310	9.53×10^{-5}
GO:BP	cellular metabolic process	1.77	482	3.42×10^{-45}
GO:BP	cellular process	1.10	827	3.50×10^{-7}
GO:BP	cellular respiration	4.50	29	2.81×10^{-9}
GO:BP	cellular response to stress	1.79	90	5.95×10^{-6}
GO:BP	chaperone cofactor-dependent protein refolding	7.84	9	3.37×10^{-4}
GO:BP	chaperone-mediated protein folding	5.03	11	1.45×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	chromosome organization	2.15	38	6.51×10^{-4}
GO:BP	cytoplasmic translation	9.15	27	7.21×10^{-17}
GO:BP	cytoplasmic translational initiation	7.48	9	4.85×10^{-4}
GO:BP	double-strand break repair via break-induced replication	7.32	6	2.28×10^{-2}
GO:BP	electron transport chain	5.08	20	5.49×10^{-7}
GO:BP	energy derivation by oxidation of organic compounds	3.00	30	1.49×10^{-5}
GO:BP	establishment of RNA localization	3.70	18	2.32×10^{-4}
GO:BP	establishment of protein localization to mitochondrion	3.09	12	3.30×10^{-2}
GO:BP	establishment of protein localization to organelle	1.91	35	1.13×10^{-2}
GO:BP	formation of cytoplasmic translation initiation complex	11.64	7	3.80×10^{-4}
GO:BP	gene expression	3.07	289	4.61×10^{-70}
GO:BP	generation of precursor metabolites and energy	2.77	40	1.13×10^{-6}
GO:BP	immunoglobulin production involved in immunoglobulin-mediated immune response	5.82	7	2.28×10^{-2}
GO:BP	import into nucleus	2.69	15	3.01×10^{-2}
GO:BP	import into the mitochondrion	3.92	12	5.82×10^{-3}
GO:BP	isotype switching	9.15	5	3.13×10^{-2}
GO:BP	mRNA metabolic process	2.21	47	4.07×10^{-5}
GO:BP	mRNA processing	2.50	39	2.09×10^{-5}
GO:BP	mRNA splicing, via spliceosome	2.70	28	2.22×10^{-4}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	macromolecule biosynthetic process	2.69	313	8.24×10^{-63}
GO:BP	macromolecule metabolic process	1.79	466	6.20×10^{-45}
GO:BP	macromolecule methylation	2.81	14	3.04×10^{-2}
GO:BP	maturation of 5.8S rRNA	6.79	13	1.21×10^{-5}
GO:BP	maturation of LSU-rRNA	8.13	16	1.88×10^{-8}
GO:BP	maturation of LSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)	5.57	7	2.86×10^{-2}
GO:BP	maturation of SSU-rRNA	6.22	17	4.15×10^{-7}
GO:BP	maturation of SSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)	6.27	12	8.09×10^{-5}
GO:BP	metabolic process	1.57	618	9.00×10^{-47}
GO:BP	mitochondrial ATP synthesis coupled electron transport	6.10	17	5.49×10^{-7}
GO:BP	mitochondrial DNA metabolic process	9.85	7	1.15×10^{-3}
GO:BP	mitochondrial DNA replication	9.15	5	3.13×10^{-2}
GO:BP	mitochondrial electron transport, NADH to ubiquinone	6.58	9	1.26×10^{-3}
GO:BP	mitochondrial gene expression	2.93	12	4.86×10^{-2}
GO:BP	mitochondrial genome maintenance	6.10	7	1.85×10^{-2}
GO:BP	mitochondrial respiratory chain complex I assembly	4.57	8	3.13×10^{-2}
GO:BP	mitochondrial transmembrane transport	3.70	18	2.32×10^{-4}
GO:BP	mitochondrial transport	3.47	22	6.14×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	mitochondrion organization	2.51	39	1.97×10^{-5}
GO:BP	mitotic cell cycle	1.69	45	2.01×10^{-2}
GO:BP	mitotic cell cycle checkpoint signaling	3.13	13	2.10×10^{-2}
GO:BP	mitotic cell cycle process	1.71	36	4.39×10^{-2}
GO:BP	negative regulation of DNA metabolic process	3.43	12	1.71×10^{-2}
GO:BP	negative regulation of cell cycle	2.19	24	1.71×10^{-2}
GO:BP	negative regulation of cell cycle phase transition	2.49	20	1.22×10^{-2}
GO:BP	negative regulation of cell cycle process	2.26	21	2.46×10^{-2}
GO:BP	non-membrane-bounded organelle assembly	1.99	30	1.66×10^{-2}
GO:BP	nuclear DNA replication	6.33	9	1.67×10^{-3}
GO:BP	nuclear export	3.33	20	3.06×10^{-4}
GO:BP	nuclear transport	2.79	30	5.99×10^{-5}
GO:BP	nucleic acid biosynthetic process	2.57	142	1.38×10^{-23}
GO:BP	nucleic acid metabolic process	2.56	226	1.57×10^{-39}
GO:BP	nucleic acid transport	3.70	18	2.32×10^{-4}
GO:BP	nucleobase-containing compound biosynthetic process	2.57	178	2.25×10^{-30}
GO:BP	nucleobase-containing compound metabolic process	2.41	276	9.74×10^{-45}
GO:BP	nucleobase-containing compound transport	3.22	22	1.83×10^{-4}
GO:BP	nucleobase-containing small molecule metabolic process	2.04	53	7.60×10^{-5}
GO:BP	nucleocytoplasmic transport	2.79	30	5.99×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	nucleoside monophosphate biosynthetic process	5.12	14	9.79×10^{-5}
GO:BP	nucleoside monophosphate metabolic process	4.57	14	3.08×10^{-4}
GO:BP	nucleoside phosphate biosynthetic process	2.50	31	2.80×10^{-4}
GO:BP	nucleoside phosphate metabolic process	2.00	46	4.85×10^{-4}
GO:BP	nucleoside triphosphate biosynthetic process	5.23	16	1.41×10^{-5}
GO:BP	nucleoside triphosphate metabolic process	3.95	27	2.73×10^{-7}
GO:BP	nucleotide biosynthetic process	2.53	31	2.28×10^{-4}
GO:BP	nucleotide metabolic process	2.10	46	1.68×10^{-4}
GO:BP	organelle organization	1.23	174	4.94×10^{-2}
GO:BP	organonitrogen compound biosynthetic process	2.74	170	3.69×10^{-32}
GO:BP	organonitrogen compound metabolic process	1.50	331	2.81×10^{-14}
GO:BP	oxidative phosphorylation	6.29	22	1.17×10^{-9}
GO:BP	positive regulation of gene expression	1.74	37	3.13×10^{-2}
GO:BP	positive regulation of signal transduction by p53 class mediator	13.06	5	8.24×10^{-3}
GO:BP	positive regulation of translation	3.59	11	2.11×10^{-2}
GO:BP	primary metabolic process	1.63	556	1.30×10^{-44}
GO:BP	protein folding	4.00	35	6.76×10^{-10}
GO:BP	protein import into nucleus	2.80	15	2.20×10^{-2}
GO:BP	protein localization to mitochondrion	3.01	12	4.05×10^{-2}
GO:BP	protein localization to nucleus	2.49	17	3.01×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	protein maturation	2.16	42	2.32×10^{-4}
GO:BP	protein metabolic process	1.52	261	2.39×10^{-11}
GO:BP	protein stabilization	3.85	16	4.94×10^{-4}
GO:BP	protein targeting	2.11	22	3.80×10^{-2}
GO:BP	protein targeting to mitochondrion	3.66	12	1.06×10^{-2}
GO:BP	protein-RNA complex assembly	5.76	40	3.24×10^{-17}
GO:BP	protein-RNA complex organization	5.54	40	1.35×10^{-16}
GO:BP	protein-containing complex assembly	2.19	83	2.81×10^{-9}
GO:BP	protein-containing complex organization	1.95	112	1.17×10^{-9}
GO:BP	proton motive force-driven ATP synthesis	6.86	9	9.28×10^{-4}
GO:BP	purine nucleoside monophosphate biosynthetic process	4.57	8	3.13×10^{-2}
GO:BP	purine nucleoside triphosphate biosynthetic process	4.91	11	1.78×10^{-3}
GO:BP	purine nucleoside triphosphate metabolic process	3.69	21	4.37×10^{-5}
GO:BP	purine nucleotide metabolic process	1.76	33	4.40×10^{-2}
GO:BP	purine ribonucleoside monophosphate biosynthetic process	4.57	8	3.13×10^{-2}
GO:BP	purine ribonucleoside triphosphate biosynthetic process	5.03	11	1.45×10^{-3}
GO:BP	purine ribonucleoside triphosphate metabolic process	3.88	21	2.09×10^{-5}
GO:BP	purine ribonucleotide metabolic process	1.94	29	2.52×10^{-2}
GO:BP	rRNA metabolic process	5.57	63	4.00×10^{-27}
GO:BP	rRNA modification	5.14	9	7.66×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	rRNA processing	5.97	62	1.61×10^{-28}
GO:BP	regulation of DNA metabolic process	2.92	30	2.42×10^{-5}
GO:BP	regulation of DNA replication	6.23	16	1.18×10^{-6}
GO:BP	regulation of DNA strand elongation	8.31	5	4.39×10^{-2}
GO:BP	regulation of DNA-templated DNA replication	7.84	6	1.71×10^{-2}
GO:BP	regulation of G2/M transition of mitotic cell cycle	3.41	11	2.94×10^{-2}
GO:BP	regulation of apoptotic process	1.58	46	4.77×10^{-2}
GO:BP	regulation of apoptotic signaling pathway	2.29	20	2.86×10^{-2}
GO:BP	regulation of cell cycle	1.95	55	1.68×10^{-4}
GO:BP	regulation of cell cycle phase transition	2.42	30	6.83×10^{-4}
GO:BP	regulation of cell cycle process	2.18	44	1.16×10^{-4}
GO:BP	regulation of protein stability	3.41	19	3.76×10^{-4}
GO:BP	regulation of signal transduction by p53 class mediator	6.10	7	1.85×10^{-2}
GO:BP	regulation of translation	2.26	20	3.13×10^{-2}
GO:BP	respiratory electron transport chain	4.99	18	4.43×10^{-6}
GO:BP	response to stress	1.45	123	9.23×10^{-4}
GO:BP	ribonucleoprotein complex biogenesis	5.52	108	1.71×10^{-47}
GO:BP	ribonucleoside monophosphate biosynthetic process	4.88	12	8.23×10^{-4}
GO:BP	ribonucleoside monophosphate metabolic process	4.39	12	2.07×10^{-3}
GO:BP	ribonucleoside triphosphate biosynthetic process	5.45	14	5.12×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	ribonucleoside triphosphate metabolic process	4.14	24	8.49×10^{-7}
GO:BP	ribonucleotide biosynthetic process	2.42	22	8.55×10^{-3}
GO:BP	ribonucleotide metabolic process	2.12	34	2.14×10^{-3}
GO:BP	ribose phosphate biosynthetic process	2.55	24	2.07×10^{-3}
GO:BP	ribose phosphate metabolic process	2.19	36	7.72×10^{-4}
GO:BP	ribosomal large subunit assembly	10.16	10	9.32×10^{-6}
GO:BP	ribosomal large subunit biogenesis	8.18	34	8.01×10^{-20}
GO:BP	ribosomal small subunit assembly	8.54	7	2.70×10^{-3}
GO:BP	ribosomal small subunit biogenesis	7.36	31	2.14×10^{-16}
GO:BP	ribosome assembly	8.13	20	6.51×10^{-11}
GO:BP	ribosome biogenesis	5.80	86	2.32×10^{-39}
GO:BP	small molecule metabolic process	1.50	106	8.46×10^{-4}
GO:BP	somatic diversification of immunoglobulins involved in immune response	9.15	5	3.13×10^{-2}
GO:BP	somatic recombination of immunoglobulin genes involved in immune response	9.15	5	3.13×10^{-2}
GO:BP	tRNA aminoacylation	3.41	11	2.94×10^{-2}
GO:BP	tRNA metabolic process	2.81	31	3.49×10^{-5}
GO:BP	tRNA transport	14.63	4	3.24×10^{-2}
GO:BP	telomere maintenance	3.35	11	3.13×10^{-2}
GO:BP	telomere organization	3.19	11	4.28×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	translation	6.31	110	7.08×10^{-55}
GO:BP	translational elongation	4.30	8	4.31×10^{-2}
GO:BP	translational initiation	5.78	12	1.79×10^{-4}
GO:BP	viral gene expression	18.29	5	1.45×10^{-3}
GO:BP	viral translation	18.29	4	1.71×10^{-2}
Cellular Component				
GO:CC	90S preribosome	8.05	20	1.61×10^{-11}
GO:CC	Arp2/3 protein complex	7.89	5	2.33×10^{-2}
GO:CC	Ctf18 RFC-like complex	15.77	5	1.30×10^{-3}
GO:CC	DNA replication preinitiation complex	9.46	5	1.18×10^{-2}
GO:CC	INO80-type complex	5.68	6	2.68×10^{-2}
GO:CC	Ino80 complex	7.89	5	2.33×10^{-2}
GO:CC	MCM complex	7.28	5	3.08×10^{-2}
GO:CC	Sm-like protein family complex	3.57	13	2.69×10^{-3}
GO:CC	U2-type prespliceosome	6.31	6	1.77×10^{-2}
GO:CC	U2-type spliceosomal complex	4.32	13	4.48×10^{-4}
GO:CC	catalytic complex	1.63	136	2.38×10^{-7}
GO:CC	catalytic step 2 spliceosome	3.40	14	2.49×10^{-3}
GO:CC	chaperonin-containing T-complex	7.36	7	2.67×10^{-3}
GO:CC	chromatin	1.64	41	2.10×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	chromosome	1.98	91	1.20×10^{-8}
GO:CC	cytochrome complex	5.10	7	1.77×10^{-2}
GO:CC	cytoplasm	1.29	685	9.10×10^{-22}
GO:CC	cytosol	2.09	237	4.82×10^{-28}
GO:CC	cytosolic large ribosomal subunit	10.51	40	1.85×10^{-29}
GO:CC	cytosolic ribosome	10.70	69	3.41×10^{-52}
GO:CC	cytosolic small ribosomal subunit	12.20	29	1.62×10^{-23}
GO:CC	endopeptidase complex	3.11	13	8.59×10^{-3}
GO:CC	eukaryotic 43S preinitiation complex	12.62	8	1.11×10^{-5}
GO:CC	eukaryotic 48S preinitiation complex	15.14	8	1.98×10^{-6}
GO:CC	eukaryotic translation initiation factor 3 complex	12.17	9	2.20×10^{-6}
GO:CC	eukaryotic translation initiation factor 3 complex, eIF3m	15.14	4	1.26×10^{-2}
GO:CC	exosome (RNase complex)	5.16	6	3.92×10^{-2}
GO:CC	fibrillar center	3.44	10	1.94×10^{-2}
GO:CC	inner mitochondrial membrane protein complex	4.50	29	6.05×10^{-10}
GO:CC	intracellular anatomical structure	1.26	951	6.67×10^{-51}
GO:CC	intracellular membrane-bounded organelle	1.39	719	2.65×10^{-36}
GO:CC	intracellular non-membrane-bounded organelle	1.85	361	2.45×10^{-34}
GO:CC	intracellular organelle	1.33	823	1.15×10^{-41}
GO:CC	intracellular organelle lumen	2.62	271	6.67×10^{-51}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	large ribosomal subunit	8.79	52	1.03×10^{-33}
GO:CC	membrane-bounded organelle	1.34	729	1.48×10^{-31}
GO:CC	membrane-enclosed lumen	2.62	271	6.67×10^{-51}
GO:CC	mitochondrial envelope	2.71	68	3.51×10^{-12}
GO:CC	mitochondrial inner membrane	3.34	50	3.51×10^{-12}
GO:CC	mitochondrial intermembrane space	4.73	9	5.04×10^{-3}
GO:CC	mitochondrial large ribosomal subunit	4.73	13	1.79×10^{-4}
GO:CC	mitochondrial matrix	3.99	42	9.69×10^{-13}
GO:CC	mitochondrial membrane	2.58	60	8.01×10^{-10}
GO:CC	mitochondrial protein-containing complex	4.57	55	9.42×10^{-20}
GO:CC	mitochondrial proton-transporting ATP synthase complex	6.08	9	8.94×10^{-4}
GO:CC	mitochondrial proton-transporting ATP synthase complex, coupling factor F(o)	6.76	5	3.92×10^{-2}
GO:CC	mitochondrial respirasome	5.16	6	3.92×10^{-2}
GO:CC	mitochondrial ribosome	5.03	21	5.82×10^{-8}
GO:CC	mitochondrial small ribosomal subunit	6.06	8	2.67×10^{-3}
GO:CC	mitochondrion	2.46	177	2.97×10^{-28}
GO:CC	non-membrane-bounded organelle	1.85	362	1.14×10^{-34}
GO:CC	nuclear chromosome	3.03	25	3.48×10^{-5}
GO:CC	nuclear envelope	2.30	34	1.90×10^{-4}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	nuclear lumen	2.49	221	4.84×10^{-37}
GO:CC	nuclear membrane	2.38	16	2.43×10^{-2}
GO:CC	nuclear pore	3.51	13	3.05×10^{-3}
GO:CC	nuclear protein-containing complex	2.21	137	2.38×10^{-17}
GO:CC	nucleolus	4.68	111	8.60×10^{-42}
GO:CC	nucleoplasm	1.94	133	3.47×10^{-12}
GO:CC	nucleus	1.53	462	4.75×10^{-26}
GO:CC	organellar large ribosomal subunit	4.73	13	1.79×10^{-4}
GO:CC	organellar ribosome	5.03	21	5.82×10^{-8}
GO:CC	organellar small ribosomal subunit	6.06	8	2.67×10^{-3}
GO:CC	organelle	1.30	829	4.21×10^{-37}
GO:CC	organelle envelope	2.56	101	8.81×10^{-17}
GO:CC	organelle envelope lumen	4.37	9	8.53×10^{-3}
GO:CC	organelle inner membrane	3.14	54	4.23×10^{-12}
GO:CC	organelle lumen	2.62	271	6.67×10^{-51}
GO:CC	organelle membrane	1.25	140	3.13×10^{-2}
GO:CC	oxidoreductase complex	4.27	14	2.31×10^{-4}
GO:CC	peptidase complex	2.70	15	1.18×10^{-2}
GO:CC	preribosome	8.06	43	6.98×10^{-26}
GO:CC	preribosome, large subunit precursor	8.60	10	1.27×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	preribosome, small subunit precursor	7.10	6	1.05×10^{-2}
GO:CC	prespliceosome	6.31	6	1.77×10^{-2}
GO:CC	protein folding chaperone complex	7.33	12	4.35×10^{-6}
GO:CC	protein-DNA complex	1.89	52	2.33×10^{-4}
GO:CC	protein-containing complex	1.73	467	5.00×10^{-40}
GO:CC	proton-transporting ATP synthase complex	5.87	9	1.16×10^{-3}
GO:CC	proton-transporting ATP synthase complex, coupling factor F(o)	7.10	6	1.05×10^{-2}
GO:CC	proton-transporting two-sector ATPase complex	3.26	10	2.66×10^{-2}
GO:CC	replication fork	4.82	13	1.51×10^{-4}
GO:CC	respirasome	5.22	8	6.59×10^{-3}
GO:CC	respiratory chain complex	5.82	8	3.33×10^{-3}
GO:CC	ribonucleoprotein complex	5.28	181	7.13×10^{-79}
GO:CC	ribosomal subunit	9.31	90	8.18×10^{-62}
GO:CC	ribosome	8.45	100	9.07×10^{-64}
GO:CC	rough endoplasmic reticulum	3.88	8	3.13×10^{-2}
GO:CC	small nuclear ribonucleoprotein complex	3.65	11	7.47×10^{-3}
GO:CC	small ribosomal subunit	10.00	37	3.59×10^{-26}
GO:CC	small-subunit processome	8.20	26	1.78×10^{-15}
GO:CC	sno(s)RNA-containing ribonucleoprotein complex	7.57	8	6.08×10^{-4}
GO:CC	spliceosomal complex	2.63	29	8.28×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	spliceosomal snRNP complex	3.86	11	4.98×10^{-3}
GO:CC	spliceosomal tri-snRNP complex	4.88	8	9.58×10^{-3}
GO:CC	translation preinitiation complex	13.10	9	1.03×10^{-6}
Molecular Function				
GO:MF	ATP hydrolysis activity	1.81	38	1.92×10^{-2}
GO:MF	ATP-dependent activity, acting on DNA	2.75	20	4.94×10^{-3}
GO:MF	ATP-dependent protein folding chaperone	4.92	15	1.11×10^{-4}
GO:MF	DNA helicase activity	4.66	13	1.02×10^{-3}
GO:MF	NADH dehydrogenase (ubiquinone) activity	9.68	6	7.91×10^{-3}
GO:MF	RNA binding	3.20	181	1.08×10^{-43}
GO:MF	catalytic activity, acting on DNA	2.37	32	1.02×10^{-3}
GO:MF	catalytic activity, acting on RNA	2.07	41	1.27×10^{-3}
GO:MF	catalytic activity, acting on a nucleic acid	2.13	71	3.41×10^{-7}
GO:MF	catalytic activity, acting on a tRNA	2.83	19	4.94×10^{-3}
GO:MF	electron transfer activity	5.16	8	2.07×10^{-2}
GO:MF	heat shock protein binding	3.45	13	1.17×10^{-2}
GO:MF	helicase activity	2.66	21	4.94×10^{-3}
GO:MF	heterocyclic compound binding	1.23	180	4.54×10^{-2}
GO:MF	hydrolase activity, acting on acid anhydrides	1.66	71	1.87×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	hydrolase activity, acting on acid anhydrides, in phosphorus-containing anhydrides	1.64	70	2.65×10^{-3}
GO:MF	identical protein binding	2.09	58	2.05×10^{-5}
GO:MF	isomerase activity	2.25	21	3.26×10^{-2}
GO:MF	mRNA binding	2.26	35	1.02×10^{-3}
GO:MF	nucleic acid binding	1.74	292	1.36×10^{-21}
GO:MF	nucleoside phosphate binding	1.24	173	3.69×10^{-2}
GO:MF	nucleotide binding	1.24	173	3.69×10^{-2}
GO:MF	organic cyclic compound binding	1.43	438	2.05×10^{-17}
GO:MF	oxidoreductase activity	1.55	68	1.17×10^{-2}
GO:MF	oxidoreductase activity, acting on NAD(P)H	4.63	11	4.18×10^{-3}
GO:MF	oxidoreductase activity, acting on NAD(P)H, quinone or similar compound as acceptor	7.74	8	2.16×10^{-3}
GO:MF	oxidoreduction-driven active transmembrane transporter activity	6.77	7	1.26×10^{-2}
GO:MF	poly(U) RNA binding	9.68	5	3.33×10^{-2}
GO:MF	protein folding chaperone	4.01	17	2.99×10^{-4}
GO:MF	protein-folding chaperone binding	3.45	13	1.17×10^{-2}
GO:MF	proton transmembrane transporter activity	2.79	17	1.23×10^{-2}
GO:MF	pyrophosphatase activity	1.65	70	2.23×10^{-3}
GO:MF	rRNA binding	7.82	21	8.21×10^{-11}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	ribonucleoprotein complex binding	3.39	17	1.91×10^{-3}
GO:MF	ribonucleoside triphosphate phosphatase activity	1.57	61	1.65×10^{-2}
GO:MF	ribosome binding	3.99	13	3.66×10^{-3}
GO:MF	single-stranded DNA binding	3.64	16	1.61×10^{-3}
GO:MF	single-stranded DNA helicase activity	7.04	8	3.73×10^{-3}
GO:MF	snoRNA binding	10.84	14	7.89×10^{-9}
GO:MF	structural constituent of nuclear pore	5.42	7	3.82×10^{-2}
GO:MF	structural constituent of ribosome	9.21	88	5.28×10^{-59}
GO:MF	structural molecule activity	2.69	109	8.75×10^{-19}
GO:MF	translation elongation factor activity	7.53	7	7.78×10^{-3}
GO:MF	translation factor activity, RNA binding	5.36	23	1.83×10^{-8}
GO:MF	translation initiation factor activity	5.58	15	2.21×10^{-5}
GO:MF	translation regulator activity	4.84	29	8.60×10^{-10}
GO:MF	translation regulator activity, nucleic acid binding	4.94	24	3.61×10^{-8}
GO:MF	unfolded protein binding	4.65	25	4.96×10^{-8}

Table 3A: Gene ontology analysis of Significantly Upregulated DEGs identified at 24-hpi

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
Biological Process				
GO:BP	ERAD pathway	6.28	14	1.77×10^{-5}
GO:BP	alcohol biosynthetic process	3.22	22	3.23×10^{-4}
GO:BP	autophagy	2.33	28	3.55×10^{-3}
GO:BP	biosynthetic process	1.21	252	1.72×10^{-2}
GO:BP	carbohydrate derivative metabolic process	1.47	84	1.58×10^{-2}
GO:BP	catabolic process	1.71	165	3.58×10^{-9}
GO:BP	cell death	1.73	42	2.50×10^{-2}
GO:BP	cellular biosynthetic process	1.24	220	1.58×10^{-2}
GO:BP	cellular catabolic process	1.76	64	1.02×10^{-3}
GO:BP	cellular homeostasis	1.64	45	3.99×10^{-2}
GO:BP	cellular lipid biosynthetic process	7.89	8	1.81×10^{-3}
GO:BP	cellular lipid metabolic process	1.60	87	1.20×10^{-3}
GO:BP	cellular localization	1.44	196	1.35×10^{-5}
GO:BP	cellular macromolecule localization	1.51	135	1.34×10^{-4}
GO:BP	cellular metabolic process	1.25	422	7.24×10^{-6}
GO:BP	cellular response to stress	1.69	105	1.87×10^{-5}
GO:BP	cellular response to topologically incorrect protein	2.87	13	4.82×10^{-2}
GO:BP	chaperone-mediated protein folding	3.70	10	3.73×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	chemical homeostasis	1.61	54	2.28×10^{-2}
GO:BP	cholesterol biosynthetic process	7.14	14	3.72×10^{-6}
GO:BP	cholesterol metabolic process	3.70	16	1.34×10^{-3}
GO:BP	cytosolic transport	2.78	16	2.09×10^{-2}
GO:BP	embryonic epithelial tube formation	3.89	10	2.70×10^{-2}
GO:BP	embryonic morphogenesis	2.03	28	2.34×10^{-2}
GO:BP	endocytosis	1.83	41	1.20×10^{-2}
GO:BP	epithelial tube formation	3.79	10	3.22×10^{-2}
GO:BP	ergosterol biosynthetic process	10.76	8	1.76×10^{-4}
GO:BP	ergosterol metabolic process	10.76	8	1.76×10^{-4}
GO:BP	establishment of localization	1.40	298	9.39×10^{-8}
GO:BP	establishment of localization in cell	1.60	145	2.78×10^{-6}
GO:BP	establishment of protein localization	1.63	100	1.68×10^{-4}
GO:BP	establishment of protein localization to organelle	1.85	42	7.96×10^{-3}
GO:BP	glycoprotein metabolic process	1.88	40	8.79×10^{-3}
GO:BP	heparan sulfate proteoglycan biosynthetic process	7.75	11	5.17×10^{-5}
GO:BP	homeostatic process	1.60	73	3.78×10^{-3}
GO:BP	intracellular monoatomic cation homeostasis	1.82	36	2.50×10^{-2}
GO:BP	intracellular monoatomic ion homeostasis	1.82	36	2.63×10^{-2}
GO:BP	intracellular pH reduction	6.12	12	1.68×10^{-4}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	intracellular protein transport	2.03	69	6.74×10^{-6}
GO:BP	intracellular signal transduction	1.39	119	8.90×10^{-3}
GO:BP	intracellular transport	1.66	118	8.14×10^{-6}
GO:BP	lipid biosynthetic process	1.77	57	2.31×10^{-3}
GO:BP	lipid metabolic process	1.50	105	1.77×10^{-3}
GO:BP	localization	1.34	322	9.94×10^{-7}
GO:BP	lysosomal lumen acidification	10.57	5	2.34×10^{-2}
GO:BP	macroautophagy	2.32	19	4.03×10^{-2}
GO:BP	macromolecule catabolic process	1.95	90	5.00×10^{-7}
GO:BP	macromolecule localization	1.48	162	4.06×10^{-5}
GO:BP	macromolecule metabolic process	1.35	435	1.22×10^{-10}
GO:BP	macromolecule modification	1.62	212	2.68×10^{-10}
GO:BP	metabolic process	1.25	609	1.89×10^{-10}
GO:BP	modification-dependent macromolecule catabolic process	1.84	49	3.12×10^{-3}
GO:BP	modification-dependent protein catabolic process	1.84	49	2.99×10^{-3}
GO:BP	monoatomic cation homeostasis	1.75	39	2.96×10^{-2}
GO:BP	monoatomic ion homeostasis	1.75	40	2.65×10^{-2}
GO:BP	morphogenesis of embryonic epithelium	3.70	11	2.25×10^{-2}
GO:BP	negative regulation of biological process	1.33	216	5.68×10^{-4}
GO:BP	negative regulation of biosynthetic process	1.56	83	3.32×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	negative regulation of cellular biosynthetic process	1.57	83	2.79×10^{-3}
GO:BP	negative regulation of cellular metabolic process	1.64	99	1.56×10^{-4}
GO:BP	negative regulation of cellular process	1.37	203	1.70×10^{-4}
GO:BP	negative regulation of cytokine production	3.15	13	2.50×10^{-2}
GO:BP	negative regulation of gene expression	2.15	54	3.27×10^{-5}
GO:BP	negative regulation of intracellular signal transduction	2.24	41	2.58×10^{-4}
GO:BP	negative regulation of macromolecule biosynthetic process	1.53	79	8.47×10^{-3}
GO:BP	negative regulation of macromolecule metabolic process	1.45	97	9.56×10^{-3}
GO:BP	negative regulation of metabolic process	1.55	112	3.27×10^{-4}
GO:BP	neural tube formation	4.03	9	4.08×10^{-2}
GO:BP	nitrogen compound transport	1.73	118	1.35×10^{-6}
GO:BP	organic hydroxy compound biosynthetic process	2.48	23	6.55×10^{-3}
GO:BP	organonitrogen compound catabolic process	1.84	100	1.01×10^{-6}
GO:BP	organonitrogen compound metabolic process	1.39	381	1.64×10^{-10}
GO:BP	peptidyl-amino acid modification	2.40	42	4.49×10^{-5}
GO:BP	peptidyl-serine modification	2.87	13	4.82×10^{-2}
GO:BP	peptidyl-threonine modification	5.28	10	3.48×10^{-3}
GO:BP	phosphate-containing compound metabolic process	1.31	158	1.20×10^{-2}
GO:BP	phospholipid biosynthetic process	2.03	24	4.91×10^{-2}
GO:BP	phospholipid metabolic process	1.71	39	4.17×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	phosphorus metabolic process	1.31	160	9.82×10^{-3}
GO:BP	phosphorylation	1.42	79	4.67×10^{-2}
GO:BP	phytosteroid biosynthetic process	9.86	10	1.77×10^{-5}
GO:BP	phytosteroid metabolic process	9.86	10	1.77×10^{-5}
GO:BP	positive regulation of apoptotic process	2.19	25	1.78×10^{-2}
GO:BP	positive regulation of biological process	1.26	249	2.53×10^{-3}
GO:BP	positive regulation of catabolic process	2.38	38	1.68×10^{-4}
GO:BP	positive regulation of cell communication	1.53	74	1.13×10^{-2}
GO:BP	positive regulation of cellular process	1.24	217	1.54×10^{-2}
GO:BP	positive regulation of intracellular signal transduction	1.61	49	3.55×10^{-2}
GO:BP	positive regulation of macromolecule metabolic process	1.32	123	3.22×10^{-2}
GO:BP	positive regulation of metabolic process	1.38	142	3.34×10^{-3}
GO:BP	positive regulation of programmed cell death	2.10	25	2.71×10^{-2}
GO:BP	positive regulation of protein catabolic process	2.64	18	1.71×10^{-2}
GO:BP	positive regulation of protein metabolic process	1.67	50	1.78×10^{-2}
GO:BP	positive regulation of response to stimulus	1.43	90	2.09×10^{-2}
GO:BP	positive regulation of signal transduction	1.64	70	3.22×10^{-3}
GO:BP	positive regulation of signaling	1.53	74	1.13×10^{-2}
GO:BP	post-translational protein modification	1.73	75	4.24×10^{-4}
GO:BP	primary metabolic process	1.32	558	9.52×10^{-14}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	process utilizing autophagic mechanism	2.33	28	3.55×10^{-3}
GO:BP	programmed cell death	1.73	42	2.50×10^{-2}
GO:BP	proteasomal protein catabolic process	2.36	42	6.24×10^{-5}
GO:BP	proteasome-mediated ubiquitin-dependent protein catabolic process	1.90	28	4.97×10^{-2}
GO:BP	protein catabolic process	2.07	71	2.36×10^{-6}
GO:BP	protein export from nucleus	4.55	8	4.17×10^{-2}
GO:BP	protein folding	2.40	26	3.87×10^{-3}
GO:BP	protein localization	1.51	135	1.30×10^{-4}
GO:BP	protein localization to organelle	1.53	57	4.32×10^{-2}
GO:BP	protein localization to vacuole	2.76	14	4.39×10^{-2}
GO:BP	protein maturation	1.70	41	3.55×10^{-2}
GO:BP	protein metabolic process	1.48	313	1.04×10^{-10}
GO:BP	protein modification by small protein conjugation	1.59	56	2.50×10^{-2}
GO:BP	protein modification by small protein conjugation or removal	1.73	73	5.68×10^{-4}
GO:BP	protein modification by small protein removal	2.51	18	2.60×10^{-2}
GO:BP	protein modification process	1.68	202	1.04×10^{-10}
GO:BP	protein phosphorylation	1.80	64	5.84×10^{-4}
GO:BP	protein transport	1.62	86	8.20×10^{-4}
GO:BP	protein ubiquitination	1.60	51	3.47×10^{-2}
GO:BP	proteoglycan biosynthetic process	3.85	13	4.99×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	proteoglycan metabolic process	3.70	15	2.31×10^{-3}
GO:BP	proteolysis	1.45	101	6.55×10^{-3}
GO:BP	proteolysis involved in protein catabolic process	1.97	63	5.57×10^{-5}
GO:BP	regulation of apoptotic process	1.62	58	1.54×10^{-2}
GO:BP	regulation of autophagy	2.31	24	1.20×10^{-2}
GO:BP	regulation of catabolic process	2.09	66	5.48×10^{-6}
GO:BP	regulation of cell communication	1.33	158	7.93×10^{-3}
GO:BP	regulation of cellular catabolic process	2.07	25	3.36×10^{-2}
GO:BP	regulation of cellular metabolic process	1.26	290	8.57×10^{-4}
GO:BP	regulation of cellular pH	3.19	19	1.47×10^{-3}
GO:BP	regulation of cytokine production	2.27	32	2.03×10^{-3}
GO:BP	regulation of cytoplasmic pattern recognition receptor signaling pathway	3.89	10	2.70×10^{-2}
GO:BP	regulation of defense response	1.88	34	2.33×10^{-2}
GO:BP	regulation of intracellular pH	3.13	18	2.85×10^{-3}
GO:BP	regulation of intracellular signal transduction	1.60	99	3.58×10^{-4}
GO:BP	regulation of lysosomal lumen pH	8.07	6	1.86×10^{-2}
GO:BP	regulation of macromolecule metabolic process	1.22	291	3.83×10^{-3}
GO:BP	regulation of metabolic process	1.27	329	6.35×10^{-5}
GO:BP	regulation of pH	2.99	19	3.08×10^{-3}
GO:BP	regulation of primary metabolic process	1.20	262	2.50×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	regulation of programmed cell death	1.58	59	2.12×10^{-2}
GO:BP	regulation of proteasomal protein catabolic process	2.60	16	3.55×10^{-2}
GO:BP	regulation of protein catabolic process	2.47	27	2.19×10^{-3}
GO:BP	regulation of protein metabolic process	1.60	85	1.41×10^{-3}
GO:BP	regulation of proteolysis involved in protein catabolic process	2.78	19	6.74×10^{-3}
GO:BP	regulation of response to stimulus	1.34	179	2.19×10^{-3}
GO:BP	regulation of response to stress	1.78	58	1.81×10^{-3}
GO:BP	regulation of signal transduction	1.41	146	1.27×10^{-3}
GO:BP	regulation of signaling	1.32	158	1.06×10^{-2}
GO:BP	response to chemical	1.49	123	6.61×10^{-4}
GO:BP	response to endoplasmic reticulum stress	3.24	23	1.76×10^{-4}
GO:BP	response to nitrogen compound	1.93	40	5.15×10^{-3}
GO:BP	response to organonitrogen compound	2.06	37	3.12×10^{-3}
GO:BP	response to stress	1.42	149	8.20×10^{-4}
GO:BP	response to topologically incorrect protein	3.00	15	1.61×10^{-2}
GO:BP	secondary alcohol biosynthetic process	7.40	17	6.60×10^{-8}
GO:BP	secondary alcohol metabolic process	3.50	18	8.20×10^{-4}
GO:BP	small molecule biosynthetic process	1.72	44	2.28×10^{-2}
GO:BP	steroid biosynthetic process	3.65	20	1.68×10^{-4}
GO:BP	steroid metabolic process	2.50	24	4.19×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	sterol biosynthetic process	6.99	17	1.44×10^{-7}
GO:BP	sterol metabolic process	3.60	19	3.27×10^{-4}
GO:BP	sulfur compound biosynthetic process	2.75	18	1.13×10^{-2}
GO:BP	sulfur compound metabolic process	1.95	30	2.66×10^{-2}
GO:BP	tissue morphogenesis	2.10	24	3.43×10^{-2}
GO:BP	transport	1.41	283	9.39×10^{-8}
GO:BP	ubiquitin-dependent protein catabolic process	1.89	49	1.89×10^{-3}
GO:BP	vacuolar acidification	6.16	10	1.18×10^{-3}
GO:BP	vacuolar transport	2.47	23	7.10×10^{-3}
GO:BP	vacuole organization	2.24	20	4.33×10^{-2}
GO:BP	vesicle organization	1.89	29	4.52×10^{-2}
GO:BP	vesicle-mediated transport	1.66	119	8.14×10^{-6}
Cellular Component				
GO:CC	ATPase complex	2.46	19	1.18×10^{-2}
GO:CC	ATPase dependent transmembrane transport complex	3.96	11	6.39×10^{-3}
GO:CC	Golgi apparatus	1.68	105	4.83×10^{-6}
GO:CC	Golgi apparatus subcompartment	1.83	28	4.19×10^{-2}
GO:CC	Golgi cisterna	3.11	12	2.23×10^{-2}
GO:CC	Golgi membrane	2.28	45	1.13×10^{-5}
GO:CC	bounding membrane of organelle	1.89	141	1.37×10^{-11}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	catalytic complex	1.30	130	2.23×10^{-2}
GO:CC	cation-transporting ATPase complex	4.14	11	4.36×10^{-3}
GO:CC	clathrin-coated vesicle	2.72	16	1.23×10^{-2}
GO:CC	coated vesicle	2.38	26	1.90×10^{-3}
GO:CC	cytoplasm	1.30	827	6.20×10^{-27}
GO:CC	cytoplasmic vesicle	1.65	128	5.29×10^{-7}
GO:CC	cytoplasmic vesicle membrane	1.85	62	9.07×10^{-5}
GO:CC	cytosol	1.58	214	4.32×10^{-10}
GO:CC	early endosome	1.94	28	2.07×10^{-2}
GO:CC	endocytic vesicle	2.26	18	3.61×10^{-2}
GO:CC	endomembrane system	1.68	315	1.99×10^{-20}
GO:CC	endoplasmic reticulum	1.88	145	8.54×10^{-12}
GO:CC	endoplasmic reticulum membrane	1.98	84	1.17×10^{-7}
GO:CC	endoplasmic reticulum subcompartment	1.98	85	9.98×10^{-8}
GO:CC	endosome	1.81	71	3.79×10^{-5}
GO:CC	endosome membrane	2.16	37	4.17×10^{-4}
GO:CC	intracellular anatomical structure	1.16	1,043	5.44×10^{-20}
GO:CC	intracellular membrane-bounded organelle	1.30	805	3.83×10^{-25}
GO:CC	intracellular organelle	1.19	881	2.20×10^{-16}
GO:CC	intracellular organelle lumen	1.53	190	5.31×10^{-8}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	intracellular protein-containing complex	1.47	79	1.13×10^{-2}
GO:CC	intracellular vesicle	1.63	128	1.01×10^{-6}
GO:CC	lysosomal membrane	3.03	32	1.47×10^{-6}
GO:CC	lysosome	2.62	52	2.58×10^{-8}
GO:CC	lytic vacuole	2.59	52	3.79×10^{-8}
GO:CC	lytic vacuole membrane	3.03	32	1.47×10^{-6}
GO:CC	membrane	1.12	612	2.43×10^{-3}
GO:CC	membrane microdomain	2.66	16	1.48×10^{-2}
GO:CC	membrane raft	2.69	16	1.35×10^{-2}
GO:CC	membrane-bounded organelle	1.26	822	4.43×10^{-22}
GO:CC	membrane-enclosed lumen	1.53	190	5.31×10^{-8}
GO:CC	nuclear body	1.81	33	2.23×10^{-2}
GO:CC	nuclear lumen	1.55	164	4.35×10^{-7}
GO:CC	nuclear outer membrane-endoplasmic reticulum membrane network	1.94	84	2.90×10^{-7}
GO:CC	nucleolus	1.62	46	2.25×10^{-2}
GO:CC	nucleoplasm	1.67	137	9.98×10^{-8}
GO:CC	nucleus	1.24	447	1.08×10^{-6}
GO:CC	organelle	1.17	893	9.33×10^{-14}
GO:CC	organelle lumen	1.53	190	5.31×10^{-8}
GO:CC	organelle membrane	1.82	244	1.57×10^{-19}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	organelle subcompartment	1.93	111	1.94×10^{-9}
GO:CC	perinuclear region of cytoplasm	2.47	34	6.02×10^{-5}
GO:CC	protein-containing complex	1.13	366	4.13×10^{-2}
GO:CC	proton-transporting V-type ATPase complex	4.52	8	2.23×10^{-2}
GO:CC	vacuolar membrane	2.73	39	9.21×10^{-7}
GO:CC	vacuolar proton-transporting V-type ATPase complex	6.33	8	2.95×10^{-3}
GO:CC	vacuole	2.48	64	2.52×10^{-9}
GO:CC	vesicle	1.56	136	3.62×10^{-6}
GO:CC	vesicle membrane	1.94	66	9.55×10^{-6}
Molecular Function				
GO:MF	acyltransferase activity	1.63	72	7.45×10^{-3}
GO:MF	binding	1.07	924	1.02×10^{-2}
GO:MF	catalytic activity	1.16	522	1.47×10^{-3}
GO:MF	catalytic activity, acting on a protein	1.28	230	4.88×10^{-3}
GO:MF	enzyme binding	1.95	120	2.10×10^{-9}
GO:MF	identical protein binding	2.17	75	2.02×10^{-7}
GO:MF	kinase binding	2.08	45	1.47×10^{-3}
GO:MF	lipid binding	1.57	62	4.54×10^{-2}
GO:MF	manganese ion binding	4.71	10	2.12×10^{-2}
GO:MF	misfolded protein binding	7.40	10	9.76×10^{-4}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	protein binding	1.24	551	4.61×10^{-8}
GO:MF	protein domain specific binding	2.12	28	3.51×10^{-2}
GO:MF	protein kinase binding	2.04	40	4.88×10^{-3}
GO:MF	steroid binding	3.20	14	3.61×10^{-2}
GO:MF	transferase activity	1.30	233	2.67×10^{-3}
GO:MF	ubiquitin-like protein ligase binding	2.50	19	4.88×10^{-2}

Table 3B: Gene ontology analysis of Significantly Downregulated DEGs identified at 24-hpi

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
Biological Process				
GO:BP	'de novo' IMP biosynthetic process	8.46	5	3.63×10^{-2}
GO:BP	'de novo' XMP biosynthetic process	13.53	4	3.42×10^{-2}
GO:BP	2'-deoxyribonucleotide biosynthetic process	7.89	7	3.48×10^{-3}
GO:BP	2'-deoxyribonucleotide metabolic process	5.26	7	3.23×10^{-2}
GO:BP	ADP catabolic process	3.54	11	2.13×10^{-2}
GO:BP	ADP metabolic process	3.46	11	2.38×10^{-2}
GO:BP	ATP biosynthetic process	5.95	11	2.49×10^{-4}
GO:BP	ATP metabolic process	3.84	23	4.27×10^{-6}
GO:BP	ATP synthesis coupled electron transport	6.26	25	1.40×10^{-11}
GO:BP	DNA damage response	2.04	81	1.12×10^{-7}
GO:BP	DNA integrity checkpoint signaling	2.95	17	5.00×10^{-3}
GO:BP	DNA metabolic process	2.27	92	2.37×10^{-11}
GO:BP	DNA recombination	2.13	32	3.19×10^{-3}
GO:BP	DNA repair	2.28	66	5.27×10^{-8}
GO:BP	DNA replication	3.29	35	9.38×10^{-8}
GO:BP	DNA-templated DNA replication	3.44	29	8.52×10^{-7}
GO:BP	DNA-templated DNA replication maintenance of fidelity	3.69	9	4.84×10^{-2}
GO:BP	GMP biosynthetic process	6.76	6	2.73×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	L-amino acid biosynthetic process	3.09	13	2.14×10^{-2}
GO:BP	L-amino acid metabolic process	2.46	28	9.12×10^{-4}
GO:BP	RNA biosynthetic process	1.71	122	4.31×10^{-7}
GO:BP	RNA metabolic process	1.89	156	1.03×10^{-12}
GO:BP	RNA processing	1.85	99	2.56×10^{-7}
GO:BP	RNA splicing	2.25	40	1.31×10^{-4}
GO:BP	RNA splicing, via transesterification reactions	2.42	34	1.68×10^{-4}
GO:BP	RNA splicing, via transesterification reactions with bulged adenosine as nucleophile	2.42	34	1.68×10^{-4}
GO:BP	XMP biosynthetic process	13.53	4	3.42×10^{-2}
GO:BP	XMP metabolic process	13.53	4	3.42×10^{-2}
GO:BP	aerobic electron transport chain	6.35	23	9.80×10^{-11}
GO:BP	aerobic respiration	5.33	41	7.30×10^{-17}
GO:BP	alpha-amino acid biosynthetic process	3.05	14	1.48×10^{-2}
GO:BP	alpha-amino acid metabolic process	2.16	29	5.33×10^{-3}
GO:BP	amino acid activation	4.80	22	1.38×10^{-7}
GO:BP	amino acid metabolic process	2.61	52	4.48×10^{-8}
GO:BP	biosynthetic process	1.78	404	6.23×10^{-33}
GO:BP	carbohydrate derivative metabolic process	1.42	89	1.91×10^{-2}
GO:BP	carboxylic acid metabolic process	1.76	78	6.39×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	catabolic process	1.30	138	2.31×10^{-2}
GO:BP	cell cycle	2.28	145	5.63×10^{-19}
GO:BP	cell cycle DNA replication	4.68	9	1.22×10^{-2}
GO:BP	cell cycle checkpoint signaling	3.05	23	1.96×10^{-4}
GO:BP	cell cycle phase transition	2.83	14	2.68×10^{-2}
GO:BP	cell cycle process	2.35	121	1.51×10^{-16}
GO:BP	cell division	2.28	40	9.83×10^{-5}
GO:BP	cellular biosynthetic process	1.83	356	7.55×10^{-31}
GO:BP	cellular component assembly	1.51	179	8.36×10^{-7}
GO:BP	cellular component biogenesis	1.61	217	5.62×10^{-11}
GO:BP	cellular component disassembly	2.06	26	2.14×10^{-2}
GO:BP	cellular component organization	1.20	381	8.36×10^{-4}
GO:BP	cellular component organization or biogenesis	1.25	417	3.54×10^{-6}
GO:BP	cellular metabolic process	1.55	571	2.12×10^{-32}
GO:BP	cellular modified amino acid metabolic process	2.19	21	3.21×10^{-2}
GO:BP	cellular process	1.10	1,121	1.42×10^{-10}
GO:BP	cellular respiration	4.93	43	2.46×10^{-16}
GO:BP	cellular response to stress	1.74	118	2.81×10^{-7}
GO:BP	centromere complex assembly	4.92	8	2.09×10^{-2}
GO:BP	chromatin organization	2.10	51	4.41×10^{-5}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	chromatin remodeling	2.07	37	1.70×10^{-3}
GO:BP	chromosome organization	2.47	59	2.36×10^{-8}
GO:BP	chromosome segregation	2.49	47	1.02×10^{-6}
GO:BP	cytoplasmic translation	8.27	33	2.36×10^{-20}
GO:BP	cytoplasmic translational initiation	7.38	12	7.93×10^{-6}
GO:BP	deoxyribonucleotide biosynthetic process	6.37	8	3.90×10^{-3}
GO:BP	deoxyribonucleotide metabolic process	4.51	8	3.17×10^{-2}
GO:BP	deoxyribose phosphate biosynthetic process	7.89	7	3.48×10^{-3}
GO:BP	deoxyribose phosphate metabolic process	4.98	7	4.10×10^{-2}
GO:BP	double-strand break repair	2.15	31	3.48×10^{-3}
GO:BP	double-strand break repair via homologous recombination	2.29	20	2.53×10^{-2}
GO:BP	electron transport chain	5.64	30	1.03×10^{-12}
GO:BP	energy derivation by oxidation of organic compounds	3.40	46	5.23×10^{-11}
GO:BP	fatty acid beta-oxidation	3.57	14	3.37×10^{-3}
GO:BP	fatty acid oxidation	3.27	14	7.85×10^{-3}
GO:BP	formation of cytoplasmic translation initiation complex	9.84	8	1.51×10^{-4}
GO:BP	formation of translation preinitiation complex	9.66	5	2.30×10^{-2}
GO:BP	gene expression	2.26	288	1.04×10^{-39}
GO:BP	generation of precursor metabolites and energy	3.07	60	1.49×10^{-12}
GO:BP	glycolytic process	3.63	11	1.81×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	import into the mitochondrion	3.14	13	1.89×10^{-2}
GO:BP	lipid oxidation	3.21	14	9.33×10^{-3}
GO:BP	mRNA metabolic process	1.95	56	1.28×10^{-4}
GO:BP	mRNA processing	2.14	45	1.28×10^{-4}
GO:BP	mRNA splicing, via spliceosome	2.42	34	1.68×10^{-4}
GO:BP	macromolecule biosynthetic process	2.03	319	5.47×10^{-35}
GO:BP	macromolecule catabolic process	1.43	72	4.46×10^{-2}
GO:BP	macromolecule metabolic process	1.54	540	3.53×10^{-29}
GO:BP	maturation of LSU-rRNA	3.76	10	2.42×10^{-2}
GO:BP	meiosis I cell cycle process	2.43	16	4.50×10^{-2}
GO:BP	meiotic cell cycle	2.42	29	8.58×10^{-4}
GO:BP	meiotic cell cycle process	2.71	28	1.68×10^{-4}
GO:BP	meiotic nuclear division	2.71	24	8.57×10^{-4}
GO:BP	metabolic process	1.43	761	3.14×10^{-36}
GO:BP	microtubule cytoskeleton organization involved in mitosis	2.67	28	2.13×10^{-4}
GO:BP	microtubule-based process	1.44	85	1.63×10^{-2}
GO:BP	mitochondrial ATP synthesis coupled electron transport	6.37	24	2.94×10^{-11}
GO:BP	mitochondrial electron transport, NADH to ubiquinone	5.41	10	1.70×10^{-3}
GO:BP	mitochondrial electron transport, succinate to ubiquinone	13.53	4	3.42×10^{-2}
GO:BP	mitochondrial gene expression	3.07	17	3.37×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	mitochondrial translation	3.64	14	2.78×10^{-3}
GO:BP	mitochondrial transmembrane transport	2.89	19	2.75×10^{-3}
GO:BP	mitochondrial transport	2.68	23	1.42×10^{-3}
GO:BP	mitochondrion organization	1.81	38	1.58×10^{-2}
GO:BP	mitotic DNA integrity checkpoint signaling	3.06	12	3.38×10^{-2}
GO:BP	mitotic cell cycle	2.64	95	5.25×10^{-16}
GO:BP	mitotic cell cycle checkpoint signaling	3.03	17	3.78×10^{-3}
GO:BP	mitotic cell cycle phase transition	2.87	14	2.38×10^{-2}
GO:BP	mitotic cell cycle process	2.68	76	9.49×10^{-13}
GO:BP	mitotic nuclear division	2.82	20	2.41×10^{-3}
GO:BP	mitotic sister chromatid segregation	2.92	19	2.39×10^{-3}
GO:BP	mitotic spindle organization	2.76	22	1.43×10^{-3}
GO:BP	negative regulation of cell cycle	2.03	30	1.10×10^{-2}
GO:BP	negative regulation of cell cycle phase transition	2.30	25	6.05×10^{-3}
GO:BP	negative regulation of cell cycle process	2.15	27	9.53×10^{-3}
GO:BP	non-membrane-bounded organelle assembly	2.40	49	1.65×10^{-6}
GO:BP	nuclear DNA replication	4.68	9	1.22×10^{-2}
GO:BP	nuclear chromosome segregation	2.47	32	2.06×10^{-4}
GO:BP	nuclear division	2.71	42	6.47×10^{-7}
GO:BP	nucleic acid biosynthetic process	1.75	131	2.42×10^{-8}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	nucleic acid metabolic process	2.02	241	3.92×10^{-25}
GO:BP	nucleobase-containing compound biosynthetic process	1.80	168	6.72×10^{-12}
GO:BP	nucleobase-containing compound catabolic process	1.84	32	2.83×10^{-2}
GO:BP	nucleobase-containing compound metabolic process	1.99	308	3.62×10^{-32}
GO:BP	nucleobase-containing small molecule metabolic process	1.91	67	2.86×10^{-5}
GO:BP	nucleoside diphosphate metabolic process	2.91	14	2.17×10^{-2}
GO:BP	nucleoside monophosphate biosynthetic process	3.25	12	2.30×10^{-2}
GO:BP	nucleoside monophosphate metabolic process	2.90	12	4.90×10^{-2}
GO:BP	nucleoside phosphate biosynthetic process	2.09	35	2.32×10^{-3}
GO:BP	nucleoside phosphate metabolic process	1.96	61	3.63×10^{-5}
GO:BP	nucleoside triphosphate biosynthetic process	4.35	18	2.11×10^{-5}
GO:BP	nucleoside triphosphate metabolic process	3.36	31	5.03×10^{-7}
GO:BP	nucleotide biosynthetic process	2.05	34	3.69×10^{-3}
GO:BP	nucleotide metabolic process	2.00	59	3.40×10^{-5}
GO:BP	organelle assembly	1.63	79	8.36×10^{-4}
GO:BP	organelle fission	2.44	42	1.11×10^{-5}
GO:BP	organelle organization	1.38	265	8.36×10^{-7}
GO:BP	organic acid metabolic process	1.70	80	1.79×10^{-4}
GO:BP	organonitrogen compound biosynthetic process	2.54	213	1.45×10^{-35}
GO:BP	organonitrogen compound metabolic process	1.46	436	4.46×10^{-17}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	oxidative phosphorylation	6.34	30	2.57×10^{-14}
GO:BP	oxoacid metabolic process	1.75	79	6.92×10^{-5}
GO:BP	peptidyl-amino acid modification	1.78	34	3.42×10^{-2}
GO:BP	peptidyl-proline modification	4.51	9	1.58×10^{-2}
GO:BP	positive regulation of apoptotic process	2.00	25	3.42×10^{-2}
GO:BP	positive regulation of signal transduction by p53 class mediator	9.66	5	2.30×10^{-2}
GO:BP	primary metabolic process	1.45	670	5.01×10^{-32}
GO:BP	protein metabolic process	1.41	327	6.60×10^{-10}
GO:BP	protein peptidyl-prolyl isomerization	6.37	8	3.90×10^{-3}
GO:BP	protein-DNA complex assembly	3.28	23	6.53×10^{-5}
GO:BP	protein-DNA complex organization	2.23	62	3.95×10^{-7}
GO:BP	protein-RNA complex assembly	4.26	40	1.03×10^{-12}
GO:BP	protein-RNA complex organization	4.20	41	8.95×10^{-13}
GO:BP	protein-containing complex assembly	2.07	106	1.19×10^{-10}
GO:BP	protein-containing complex organization	1.97	153	5.98×10^{-14}
GO:BP	proteinogenic amino acid biosynthetic process	3.09	13	2.14×10^{-2}
GO:BP	proteinogenic amino acid metabolic process	2.46	26	1.73×10^{-3}
GO:BP	proton motive force-driven ATP synthesis	6.20	11	1.68×10^{-4}
GO:BP	purine nucleoside diphosphate catabolic process	3.46	11	2.38×10^{-2}
GO:BP	purine nucleoside diphosphate metabolic process	3.61	12	1.02×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	purine nucleoside triphosphate biosynthetic process	4.29	13	1.08×10^{-3}
GO:BP	purine nucleoside triphosphate metabolic process	3.25	25	2.63×10^{-5}
GO:BP	purine nucleotide biosynthetic process	1.95	26	3.69×10^{-2}
GO:BP	purine nucleotide metabolic process	1.90	48	1.11×10^{-3}
GO:BP	purine ribonucleoside diphosphate catabolic process	3.46	11	2.38×10^{-2}
GO:BP	purine ribonucleoside diphosphate metabolic process	3.38	11	2.75×10^{-2}
GO:BP	purine ribonucleoside triphosphate biosynthetic process	4.06	12	3.66×10^{-3}
GO:BP	purine ribonucleoside triphosphate metabolic process	3.28	24	3.71×10^{-5}
GO:BP	purine ribonucleotide biosynthetic process	2.09	23	3.25×10^{-2}
GO:BP	purine ribonucleotide metabolic process	2.03	41	1.00×10^{-3}
GO:BP	purine-containing compound biosynthetic process	1.93	27	3.50×10^{-2}
GO:BP	purine-containing compound metabolic process	1.86	50	1.20×10^{-3}
GO:BP	pyridine nucleotide catabolic process	3.46	11	2.38×10^{-2}
GO:BP	pyridine-containing compound catabolic process	3.38	11	2.75×10^{-2}
GO:BP	pyruvate metabolic process	2.87	14	2.38×10^{-2}
GO:BP	rRNA metabolic process	2.48	38	2.68×10^{-5}
GO:BP	rRNA processing	2.56	36	2.63×10^{-5}
GO:BP	recombinational repair	2.24	20	3.24×10^{-2}
GO:BP	regulation of DNA metabolic process	2.01	28	1.89×10^{-2}
GO:BP	regulation of DNA replication	3.74	13	3.86×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	regulation of G2/M transition of mitotic cell cycle	3.44	15	2.78×10^{-3}
GO:BP	regulation of apoptotic process	1.53	60	2.68×10^{-2}
GO:BP	regulation of apoptotic signaling pathway	2.11	25	1.93×10^{-2}
GO:BP	regulation of cell cycle	2.05	78	2.06×10^{-7}
GO:BP	regulation of cell cycle G2/M phase transition	3.18	16	3.60×10^{-3}
GO:BP	regulation of cell cycle phase transition	2.32	39	8.76×10^{-5}
GO:BP	regulation of cell cycle process	2.13	58	5.99×10^{-6}
GO:BP	regulation of cellular response to stress	2.09	30	6.75×10^{-3}
GO:BP	regulation of chromosome organization	2.35	17	4.54×10^{-2}
GO:BP	regulation of chromosome segregation	3.10	14	1.27×10^{-2}
GO:BP	regulation of double-strand break repair	2.90	12	4.90×10^{-2}
GO:BP	regulation of metaphase/anaphase transition of cell cycle	3.76	10	2.42×10^{-2}
GO:BP	regulation of mitotic cell cycle	2.05	37	1.99×10^{-3}
GO:BP	regulation of mitotic cell cycle phase transition	2.28	26	5.00×10^{-3}
GO:BP	regulation of mitotic metaphase/anaphase transition	3.87	10	2.13×10^{-2}
GO:BP	regulation of mitotic sister chromatid separation	3.80	9	4.04×10^{-2}
GO:BP	regulation of translation	2.00	24	4.09×10^{-2}
GO:BP	respiratory electron transport chain	5.74	28	5.17×10^{-12}
GO:BP	response to stress	1.40	161	3.57×10^{-4}
GO:BP	ribonucleoprotein complex biogenesis	3.17	84	5.63×10^{-19}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	ribonucleoside diphosphate catabolic process	3.38	11	2.75×10^{-2}
GO:BP	ribonucleoside diphosphate metabolic process	3.25	12	2.30×10^{-2}
GO:BP	ribonucleoside triphosphate biosynthetic process	4.03	14	1.00×10^{-3}
GO:BP	ribonucleoside triphosphate metabolic process	3.32	26	1.07×10^{-5}
GO:BP	ribonucleotide biosynthetic process	2.04	25	2.81×10^{-2}
GO:BP	ribonucleotide metabolic process	2.03	44	5.34×10^{-4}
GO:BP	ribose phosphate biosynthetic process	2.20	28	5.00×10^{-3}
GO:BP	ribose phosphate metabolic process	2.16	48	4.64×10^{-5}
GO:BP	ribosomal large subunit biogenesis	3.74	21	2.61×10^{-5}
GO:BP	ribosomal small subunit assembly	6.31	7	1.33×10^{-2}
GO:BP	ribosomal small subunit biogenesis	3.69	21	3.07×10^{-5}
GO:BP	ribosome assembly	5.71	19	9.60×10^{-8}
GO:BP	ribosome biogenesis	2.90	58	5.23×10^{-11}
GO:BP	sexual reproduction	1.74	51	4.56×10^{-3}
GO:BP	signal transduction in response to DNA damage	2.54	15	4.38×10^{-2}
GO:BP	sister chromatid segregation	2.91	20	1.62×10^{-3}
GO:BP	small molecule metabolic process	1.54	147	7.02×10^{-6}
GO:BP	spindle organization	2.62	32	6.68×10^{-5}
GO:BP	tRNA aminoacylation	5.04	22	5.27×10^{-8}
GO:BP	tRNA aminoacylation for protein translation	4.92	20	5.03×10^{-7}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:BP	tRNA metabolic process	2.28	34	5.83×10^{-4}
GO:BP	tetrahydrofolate metabolic process	6.24	6	3.75×10^{-2}
GO:BP	translation	5.77	136	3.59×10^{-65}
GO:BP	translational elongation	4.77	12	8.57×10^{-4}
GO:BP	translational initiation	5.34	15	1.59×10^{-5}
GO:BP	tricarboxylic acid cycle	5.34	15	1.59×10^{-5}
Cellular Component				
GO:CC	Arp2/3 protein complex	6.92	6	1.02×10^{-2}
GO:CC	Sm-like protein family complex	3.21	16	1.16×10^{-3}
GO:CC	U1 snRNP	5.10	7	1.67×10^{-2}
GO:CC	U12-type spliceosomal complex	4.62	7	2.88×10^{-2}
GO:CC	U2 snRNP	5.54	8	3.72×10^{-3}
GO:CC	U2-type spliceosomal complex	3.64	15	5.33×10^{-4}
GO:CC	U4 snRNP	10.39	6	1.12×10^{-3}
GO:CC	U5 snRNP	7.55	6	6.68×10^{-3}
GO:CC	aminoacyl-tRNA synthetase multienzyme complex	8.08	7	1.12×10^{-3}
GO:CC	catalytic complex	1.64	188	1.31×10^{-10}
GO:CC	catalytic step 2 spliceosome	3.02	17	1.37×10^{-3}
GO:CC	centrosome	1.78	55	5.33×10^{-4}
GO:CC	chromatin	1.69	58	1.18×10^{-3}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	chromosomal region	2.85	41	6.00×10^{-8}
GO:CC	chromosome	2.09	131	2.56×10^{-14}
GO:CC	chromosome, centromeric region	2.94	34	7.34×10^{-7}
GO:CC	cleavage furrow	4.82	8	9.12×10^{-3}
GO:CC	condensed chromosome	2.77	36	1.29×10^{-6}
GO:CC	condensed chromosome, centromeric region	2.70	24	3.03×10^{-4}
GO:CC	cytochrome complex	5.86	11	1.01×10^{-4}
GO:CC	cytoplasm	1.28	929	7.97×10^{-28}
GO:CC	cytosol	1.95	302	3.27×10^{-30}
GO:CC	cytosolic large ribosomal subunit	8.46	44	4.02×10^{-29}
GO:CC	cytosolic ribosome	8.63	76	4.32×10^{-52}
GO:CC	cytosolic small ribosomal subunit	9.54	31	1.85×10^{-22}
GO:CC	eukaryotic 43S preinitiation complex	9.23	8	9.12×10^{-5}
GO:CC	eukaryotic 48S preinitiation complex	11.08	8	1.63×10^{-5}
GO:CC	eukaryotic translation initiation factor 3 complex	8.90	9	2.36×10^{-5}
GO:CC	eukaryotic translation initiation factor 3 complex, eIF3m	11.08	4	3.24×10^{-2}
GO:CC	inner mitochondrial membrane protein complex	3.86	34	4.75×10^{-10}
GO:CC	intracellular anatomical structure	1.23	1,265	8.08×10^{-52}
GO:CC	intracellular membrane-bounded organelle	1.28	905	4.52×10^{-26}
GO:CC	intracellular non-membrane-bounded organelle	1.72	459	4.71×10^{-35}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	intracellular organelle	1.29	1,087	1.05×10^{-42}
GO:CC	intracellular organelle lumen	1.94	275	3.97×10^{-27}
GO:CC	intracellular protein-containing complex	1.44	89	4.68×10^{-3}
GO:CC	kinetochore	2.89	24	1.01×10^{-4}
GO:CC	large ribosomal subunit	6.31	51	3.76×10^{-26}
GO:CC	membrane-bounded organelle	1.26	933	9.54×10^{-25}
GO:CC	membrane-enclosed lumen	1.94	275	3.97×10^{-27}
GO:CC	microtubule cytoskeleton	1.48	111	4.41×10^{-4}
GO:CC	microtubule organizing center	1.54	60	1.02×10^{-2}
GO:CC	mitochondrial envelope	2.13	73	3.18×10^{-8}
GO:CC	mitochondrial inner membrane	2.94	60	2.55×10^{-12}
GO:CC	mitochondrial matrix	3.48	50	4.42×10^{-13}
GO:CC	mitochondrial membrane	2.17	69	4.32×10^{-8}
GO:CC	mitochondrial protein-containing complex	3.58	59	2.73×10^{-16}
GO:CC	mitochondrial proton-transporting ATP synthase complex	6.43	13	4.00×10^{-6}
GO:CC	mitochondrial proton-transporting ATP synthase complex, coupling factor F(o)	5.93	6	2.21×10^{-2}
GO:CC	mitochondrial respirasome	5.66	9	1.12×10^{-3}
GO:CC	mitochondrial ribosome	3.68	21	9.69×10^{-6}
GO:CC	mitochondrial small ribosomal subunit	6.65	12	8.40×10^{-6}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	mitochondrion	2.39	235	6.18×10^{-36}
GO:CC	mitotic spindle	2.16	17	4.58×10^{-2}
GO:CC	non-membrane-bounded organelle	1.72	460	2.54×10^{-35}
GO:CC	nuclear chromosome	1.95	22	3.89×10^{-2}
GO:CC	nuclear lumen	1.85	224	6.28×10^{-19}
GO:CC	nuclear protein-containing complex	1.60	136	7.62×10^{-7}
GO:CC	nucleolus	2.34	76	1.62×10^{-10}
GO:CC	nucleoplasm	1.80	169	1.02×10^{-12}
GO:CC	nucleus	1.36	561	2.14×10^{-17}
GO:CC	organellar ribosome	3.68	21	9.69×10^{-6}
GO:CC	organellar small ribosomal subunit	6.65	12	8.40×10^{-6}
GO:CC	organelle	1.26	1,100	9.05×10^{-39}
GO:CC	organelle envelope	1.93	104	2.99×10^{-9}
GO:CC	organelle inner membrane	2.73	64	1.16×10^{-11}
GO:CC	organelle lumen	1.94	275	3.97×10^{-27}
GO:CC	oxidoreductase complex	4.24	19	4.39×10^{-6}
GO:CC	pICln-Sm protein complex	9.89	5	8.41×10^{-3}
GO:CC	preribosome	2.74	20	1.16×10^{-3}
GO:CC	proteasome core complex	6.23	9	5.43×10^{-4}
GO:CC	protein-DNA complex	1.72	65	3.03×10^{-4}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:CC	protein-containing complex	1.54	570	5.76×10^{-32}
GO:CC	proton-transporting ATP synthase complex	6.21	13	5.94×10^{-6}
GO:CC	proton-transporting ATP synthase complex, coupling factor F(o)	5.19	6	3.99×10^{-2}
GO:CC	proton-transporting two-sector ATPase complex	3.58	15	6.27×10^{-4}
GO:CC	replication fork	3.53	13	2.44×10^{-3}
GO:CC	respirasome	5.25	11	3.03×10^{-4}
GO:CC	respiratory chain complex	5.86	11	1.01×10^{-4}
GO:CC	respiratory chain complex IV	5.70	7	9.06×10^{-3}
GO:CC	ribonucleoprotein complex	3.78	177	5.05×10^{-54}
GO:CC	ribosomal subunit	7.04	93	1.92×10^{-53}
GO:CC	ribosome	6.68	108	4.42×10^{-59}
GO:CC	small nuclear ribonucleoprotein complex	3.64	15	5.33×10^{-4}
GO:CC	small ribosomal subunit	8.11	41	3.74×10^{-26}
GO:CC	small-subunit processome	2.77	12	3.26×10^{-2}
GO:CC	spindle	2.33	45	4.67×10^{-6}
GO:CC	spliceosomal complex	2.45	37	1.52×10^{-5}
GO:CC	spliceosomal snRNP complex	3.59	14	1.12×10^{-3}
GO:CC	spliceosomal tri-snRNP complex	4.47	10	2.72×10^{-3}
GO:CC	translation preinitiation complex	9.59	9	1.15×10^{-5}
GO:CC	tricarboxylic acid cycle heteromeric enzyme complex	6.92	5	3.76×10^{-2}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
Molecular Function				
GO:MF	ATP-dependent activity, acting on DNA	2.32	23	1.60×10^{-2}
GO:MF	NAD binding	3.78	17	5.51×10^{-4}
GO:MF	NAD ⁺ binding	6.22	7	2.33×10^{-2}
GO:MF	RNA binding	2.37	182	6.82×10^{-26}
GO:MF	aminoacyl-tRNA ligase activity	5.08	20	5.57×10^{-7}
GO:MF	binding	1.05	999	2.45×10^{-2}
GO:MF	catalytic activity	1.13	553	9.04×10^{-3}
GO:MF	catalytic activity, acting on DNA	2.07	38	2.37×10^{-3}
GO:MF	catalytic activity, acting on RNA	1.89	51	1.40×10^{-3}
GO:MF	catalytic activity, acting on a nucleic acid	1.91	87	7.53×10^{-7}
GO:MF	catalytic activity, acting on a tRNA	3.17	29	8.40×10^{-6}
GO:MF	electron transfer activity	5.21	11	1.73×10^{-3}
GO:MF	heterocyclic compound binding	1.24	247	8.80×10^{-3}
GO:MF	identical protein binding	1.77	67	5.59×10^{-4}
GO:MF	isomerase activity	2.20	28	9.04×10^{-3}
GO:MF	ligase activity	2.63	45	6.96×10^{-7}
GO:MF	ligase activity, forming carbon-oxygen bonds	5.08	20	5.57×10^{-7}
GO:MF	mRNA binding	1.99	42	2.32×10^{-3}
GO:MF	nucleic acid binding	1.49	340	2.94×10^{-13}

GO Category	GO:Term	Fold Enrichment	Number of DEGs	P-value (Adjusted)
GO:MF	nucleoside phosphate binding	1.25	236	9.04×10^{-3}
GO:MF	nucleotide binding	1.25	236	9.04×10^{-3}
GO:MF	organic cyclic compound binding	1.30	544	9.27×10^{-12}
GO:MF	oxidoreductase activity	1.61	96	3.56×10^{-4}
GO:MF	oxidoreductase activity, acting on NAD(P)H	3.71	12	1.38×10^{-2}
GO:MF	oxidoreductase activity, acting on NAD(P)H, quinone or similar compound as acceptor	5.69	8	1.42×10^{-2}
GO:MF	oxidoreductase activity, acting on the CH-NH group of donors, NAD or NADP as acceptor	6.63	7	1.60×10^{-2}
GO:MF	proton transmembrane transporter activity	2.41	20	2.45×10^{-2}
GO:MF	rRNA binding	6.56	24	3.02×10^{-11}
GO:MF	single-stranded DNA binding	2.84	17	1.33×10^{-2}
GO:MF	structural constituent of ribosome	7.30	95	1.45×10^{-55}
GO:MF	structural molecule activity	2.26	125	1.73×10^{-15}
GO:MF	translation elongation factor activity	5.53	7	4.38×10^{-2}
GO:MF	translation factor activity, RNA binding	4.28	25	2.61×10^{-7}
GO:MF	translation initiation factor activity	4.37	16	1.74×10^{-4}
GO:MF	translation regulator activity	3.92	32	1.15×10^{-8}
GO:MF	translation regulator activity, nucleic acid binding	3.93	26	5.88×10^{-7}

Table 4: Significantly Enriched KEGG Pathways from DEGs identified at 12- and 24-hpi (Results from the DAVID online resource)

Time Point	Regulation	KEGG Term	DEG Count	Fold Enrichment	P-value (Adjusted)
12-hpi	down	Ribosome	80	6.68	3.16×10^{-49}
12-hpi	down	Oxidative phosphorylation	37	3.22	1.08×10^{-8}
12-hpi	down	DNA replication	18	6.01	1.09×10^{-8}
12-hpi	down	Ribosome biogenesis in eukaryotes	27	4.03	1.09×10^{-8}
12-hpi	down	Spliceosome	30	2.50	1.25×10^{-4}
12-hpi	down	Nucleocytoplasmic transport	22	2.29	1.00×10^{-2}
12-hpi	down	Base excision repair	13	3.10	1.13×10^{-2}
12-hpi	down	Mismatch repair	9	4.29	1.13×10^{-2}
12-hpi	down	Nucleotide excision repair	14	2.86	1.29×10^{-2}
12-hpi	up	Steroid biosynthesis	10	6.14	1.65×10^{-3}
12-hpi	up	Autophagy - animal	29	2.34	2.12×10^{-3}
12-hpi	up	Cell cycle	27	2.30	3.90×10^{-3}
12-hpi	up	Influenza A	22	2.13	4.74×10^{-2}
24-hpi	down	Ribosome	88	5.54	2.81×10^{-49}
24-hpi	down	Oxidative phosphorylation	50	3.28	2.71×10^{-13}
24-hpi	down	Carbon metabolism	39	2.98	1.08×10^{-8}
24-hpi	down	Aminoacyl-tRNA biosynthesis	22	3.78	1.10×10^{-6}
24-hpi	down	Biosynthesis of amino acids	24	3.02	2.50×10^{-5}

Table 4: Significantly Enriched KEGG Pathways from DEGs identified at 12- and 24-hpi (Results from the DAVID online resource)

Time Point	Regulation	KEGG Term	DEG Count	Fold Enrichment	P-value (Adjusted)
24-hpi	down	Citrate cycle (TCA cycle)	15	4.36	2.50×10^{-5}
24-hpi	down	DNA replication	15	3.78	1.93×10^{-4}
24-hpi	down	Spliceosome	33	2.08	1.09×10^{-3}
24-hpi	down	Metabolic pathways	225	1.22	3.04×10^{-3}
24-hpi	down	Cell cycle	36	1.89	3.04×10^{-3}
24-hpi	down	Propanoate metabolism	12	3.24	7.53×10^{-3}
24-hpi	down	Fatty acid degradation	14	2.86	7.77×10^{-3}
24-hpi	down	Glycolysis / Gluconeogenesis	17	2.42	1.19×10^{-2}
24-hpi	down	One carbon pool by folate	9	3.78	1.35×10^{-2}
24-hpi	down	Nucleotide excision repair	15	2.31	3.73×10^{-2}
24-hpi	down	Pyruvate metabolism	12	2.59	4.20×10^{-2}
24-hpi	up	Steroid biosynthesis	11	5.15	1.92×10^{-3}
24-hpi	up	Lysosome	29	2.24	3.94×10^{-3}
24-hpi	up	Terpenoid backbone biosynthesis	9	4.43	1.73×10^{-2}
24-hpi	up	Glycosaminoglycan biosynthesis - heparan sulfate / heparin	10	3.90	1.73×10^{-2}
24-hpi	up	Protein processing in endoplasmic reticulum	30	1.94	1.73×10^{-2}
24-hpi	up	Autophagy - animal	30	1.85	3.19×10^{-2}

4. Discussion

THEV has a worldwide distribution, wreaking economic havoc on affected poultry farms, particularly due to its immunosuppressive trait allowing secondary opportunistic infections to devastate turkey populations (4, 6). HE in turkeys causes more economic losses than any disease caused in other birds like chicken or pheasants (4). While the current vaccine strain (VAS) has proven effective at preventing clinical HE in turkey poults, the retention of its immunosuppressive properties leaves some of the issues associated with economic losses unresolved. Elucidating the virus-host interactions leading to IMS is most pressing for not only the understanding of the viral infection and pathogenesis but also future antiviral therapy targets. Since both virulent and avirulent THEV cause IMS but the avirulent are used as vaccine, we believe that studying VAS would be more expedient for understanding THEV vaccine-induced IMS.

Only one cell line (MDTC-RP19 also known as RP19) has been found to be capable of supporting THEV replication (39). Thus, in this work, we establish the first transcriptomic profile of THEV infection in RP19 cells using paired-end RNA-seq. We attempted a multi-time point experimental design but this being the first transcriptomic study of THEV infection, we faced some difficulties, including selecting our sampling time points based on the only study of THEV gene expression kinetics (40), leading to only 12- and 24-hpi providing useful data. In total **2,343** and **3,295** DEGs were identified at 12-hpi and 24-hpi, respectively. At 12-hpi, **1,079** genes were upregulated and **1,264** genes downregulated, whereas **1,512** genes were upregulated and **1,783** genes downregulated at 24-hpi. Being a non-model organism, a significant proportion of the host (*M. gallopavo*) genes are not annotated and not recognized by the databases used for functional enrichment analysis. Thus, the obtained results are likely sub-optimal in amount of detail relative to results from well annotated and curated genomes of model organisms. The DEGs were related to multiple biological processes all potentially playing a role in THEV infection but the most relevant to our study are apoptosis (upregulated), ER stress-induced unfolded protein response (upregulated), cell maintenance processes (downregulated), and cytokine functions (deregulated). Furthermore, the RT-qPCR results validated the RNA-seq results. Collectively, this study may shed light on some significant aspects of THEV-host interactions, which may benefit further mechanistic delineation of the viral infection and induction of IMS and inform future development of anti-THEV strategies. The biological processes most relevant to THEV-induced IMS highlighted by this study are further discussed below.

Apoptosis is a key defense mechanism activated by cells in response to irreversible injury and virus infection to abrogate virus propagation. It is a formidable cellular defense network, non-specific to any virus family and therefore an important problem for any infecting virus to overcome (31–33). The adenovirus E1A proteins are strong inducers of apoptosis. They bind host pRb and p300/CBP protein, inducing p53-mediated

apoptosis, and can also sensitize infected cells to TNF α and TRAIL-induced apoptosis (34, 35). However, adenoviruses have developed multiple distinct anti-apoptotic mechanisms to counter almost all cellular pro-apoptotic programs. For example, E1A blocks its own induction of p53-dependent apoptosis and E1B proteins (E1B-19K and E1B-55K) counteract several types of apoptosis including TNF-induced apoptosis (34, 35). Despite the rich arsenal of countermeasures, transcriptomic studies of human adenovirus infections suggest a complex set of virus-host interactions where both pro- and anti-apoptotic genes are turned on contemporaneously. For example, in human adenovirus 2 infection, both pro- and anti-apoptotic BCL2 family genes were stimulated (34). Siadenoviruses including THEV are the smallest adenoviruses and therefore encode the fewest genes (10, 41). THEV encodes a mere 34 ORFs with no anti-apoptotic genes characterized (41). In agreement with these findings, in our results a strong signal indicative of apoptotic induction was observed. However, like Mastadenovirus infections, a complex relationship between pro and anti-apoptotic genes were observed. Pro-apoptotic genes such as *APAF1*, *BNIP3L*, *BMF*, *BAK1*, *RIPK1*, *FAS*, *FADD* and *ATF* were upregulated in concert with the anti-apoptotic genes *BCL2*, *BNIP2*, *BCL2A1* and *API5*. We speculate that this complex regulation is predictive of THEV possessing some anti-apoptotic genes but not sufficiently potent to thwart the cellular apoptotic response. It is also possible that using a naturally attenuated strain may account for the apparent balance in pro- and anti-apoptotic signals. Interestingly, pro-apoptotic genes in both intrinsic and extrinsic pathways were upregulated, possibly due to a concurrent stimulation of multiple apoptotic pathways or a positive feedback mechanism of one system activating the other. The specific mechanism of apoptosis induction remains elusive. Further studies designed to elucidate these fine details are warranted and would benefit future THEV therapeutics.

The ER serves many specialized functions including biosynthesis and assembly of membrane and secretory proteins, calcium storage, and biosynthesis of lipids and sterols. It is also the site of protein folding and post-translational modifications and maintains stringent quality control systems, ensuring exported proteins are correctly folded and degradation of unfolded or misfolded proteins (16, 38, 42). Disruption of ER homeostasis or ER stress leads to accumulation of incorrect proteins in the ER lumen, triggering the UPR. The UPR restores ER normality by transiently attenuating general protein synthesis, increasing the luminal folding capacity, and the degradation of misfolded proteins through the ERAD pathway or autophagy (16, 38, 42, 43). However, if incorrect luminal protein overload persists, the prolonged UPR will induce apoptosis known as ER stress-associated programmed cell death (42, 43). Many DNA and RNA viruses are reported to induce ER stress and UPR pathways during infection (16). In our results, *ATF4* and PKR-like ER protein kinase (*PERK*), key proteins in the *PERK* branch of the UPR pathway were upregulated. A myriad of ERAD pathway proteins (e.g., *VCP*, *UFD1*, *EDEM1*, *EDEM3*, *CUL1*, *UBQLN1*), ubiquitination system

proteins (e.g., *UBE2J2*, *UBE2E3*, *UBE2Z*, *UBE3A*, *UBE3B*), and heat shock family of chaperone proteins (e.g., *HSPA5*, *HSP4L*, *HSPA8*, *HSP90AA1*) all showed increased expression according to our RNA-seq data with some validated with RT-qPCR. These data strongly suggest that THEV infection triggers the ER UPR pathways leading to a massive decrease of protein synthesis and deregulation of sterol biosynthesis, and ubiquitin-mediated proteolysis, all supported by our results. As noted above, a prolonged UPR activation leads to ER stress-associated programmed cell death via genes such *ATF4* (42, 43). Thus, we suggest that the ER stress response likely plays a crucial role in THEV-induced IMS. Nonetheless, the mechanisms underlying the regulation of the UPR pathways by THEV remain to be clearly unraveled. Also, whether and how the ER stress response affects THEV infection and pathogenicity should be studied. Unsurprisingly, protein degradation was more evident at 24-hpi than at 12-hpi, reflecting the suggested two phases of UPR – Phase 1 allows the unfolded proteins time to refold without degradation and Phase 2 degrades any proteins which have failed to fold (43).

In the proposed model of THEV immunopathogenesis by Rautenschlein *et al* (**Figure 1**), while THEV directly induced cell death in infected cells, cytokines are responsible for extending cell death to bystander splenocytes (8). However, the primary cytokines (IFN- α , IFN- β , IFN- γ , TNF, IL-6, and NO) highlighted in the model were not significantly differentially expressed in our data. This may be explained by the fact that the model was proposed based on data from splenocytes of THEV-infected turkeys, which have the full complement of immune cells (T-cells, B-cells, macrophages) shown in the model and not solely from B-cell culture data as in this study. From the model, T-cells and macrophages are the principal producers of the effector cytokines; thus, there is agreement with our data that B-cells alone would poorly recapitulate the cytokine communication network. This may also explain the very few immune-associated biological processes in our data as the B-cells may require cytokines from other immune cells such as macrophages and T-cells for optimal activation. Further transcriptomic studies with splenocytes would offer a wealth of insights regarding these ideas. It also likely that cytokines may only play a dominant role in some aspects of THEV-infection such as the clinical hemorrhage of the intestines but not the associated IMS since a study using the TNF-blocking drug (thalidomide) only prevented intestinal disease, not IMS (8). While some of the upregulated cytokines and receptors in our results are positive apoptosis regulators (*TNFRSF8*, *TRAF7*), most of the cytokines are either anti-apoptotic (*TNFRSF1B*, *TRAF2*), boost host antiviral defense (*IL18*, *TNFSF4*, *PKR*, *TRAFD1*, *IFIH1*), or suppress cytokine signaling (*SOCS3*, *SOCS5*). Therefore, we speculate that a non-cytokine-mediated apoptotic process such as ER stress-associated programmed cell death is more likely to mediate direct killing of infected cells. However, whether bystander cell death occurs and if it is cytokine-mediated as suggested by Rautenschlein *et al* are important questions that can be addressed

with future transcriptomic studies in splenocytes.

By convention, the Mastadenovirus replication cycle is divided into two phases, an early and a late phase, separated by the onset of viral DNA replication (34, 35). Based on DNA microarray analysis, adenovirus type 2 (Ad2) infection has been divided into four stages. The first period is from 0 to 12-hpi during which changes in cellular gene expression are likely to be triggered by the viral entry process. Most of the deregulated genes have functions linked to inhibition of cell growth. Therefore, growth suppression is most likely the first response of the host cell to the incoming virus (34). The second period covers the time from 12 to 24-hpi and follows activation of the immediate early E1A gene, which forcibly transition cell cycle to S phase (34). While the temporal changes of host gene expression for a THEV infection have not been characterized in prior studies, our data suggest that during the first 24-hpi, cell growth was suppressed. Cell maintenance processes involving nucleic acid and proteins were downregulated according to our data. Protein synthesis-related processes including ribosome biogenesis, rRNA processing, ribosome assembly, protein folding, translational initiation, protein maturation, and others were heavily affected. Additionally, DNA and RNA synthesis, maintenance, and repair such as nucleotide biosynthesis and metabolism, double strand break repair, and DNA excision repair were also repressed. As severe DNA damage leads to DNA damage-dependent apoptosis (36) and repression of RNA and protein synthesis is also strongly associated with apoptosis (37), these inhibitions may also play a role in THEV-induced IMS. Moreover, we speculate that the ER UPR may contribute partly to the severe repression of protein synthesis as discussed above. An in-depth study of temporal changes of host gene expression during THEV infection would be invaluable in establishing if THEV follows the same pattern as Ad2.

5. Conclusions

THEV-induced IMS is a pressing concern for turkey farmers worldwide, causing substantial economic losses annually. In this study, we establish the cellular transcriptomic profile of THEV infection in turkey RP19 B-cells using paired-end RNA-seq, identifying **1,079** upregulated genes and **1,264** downregulated genes at 12-hpi and **1,512** upregulated genes and **1,783** downregulated genes at 24-hpi. Our data suggest that several biological processes and pathways including apoptosis, immune response, ER response to stress, ubiquitin-dependent proteolysis, and repression of essential cellular maintenance are significant aspects of host cell response to THEV infection. All these processes are established apoptosis inducing mechanisms; therefore, we believe that either one or synergistic interplay between multiple ones may mediate cell death of infected B-cells, leading to IMS. These findings provide the first insights into THEV-host interactions and may help advance the understanding of non-human adenoviral infection and pathogenesis, which may

⁴⁶⁷ eventually inform the development of medical countermeasures for disease prevention and treatment.

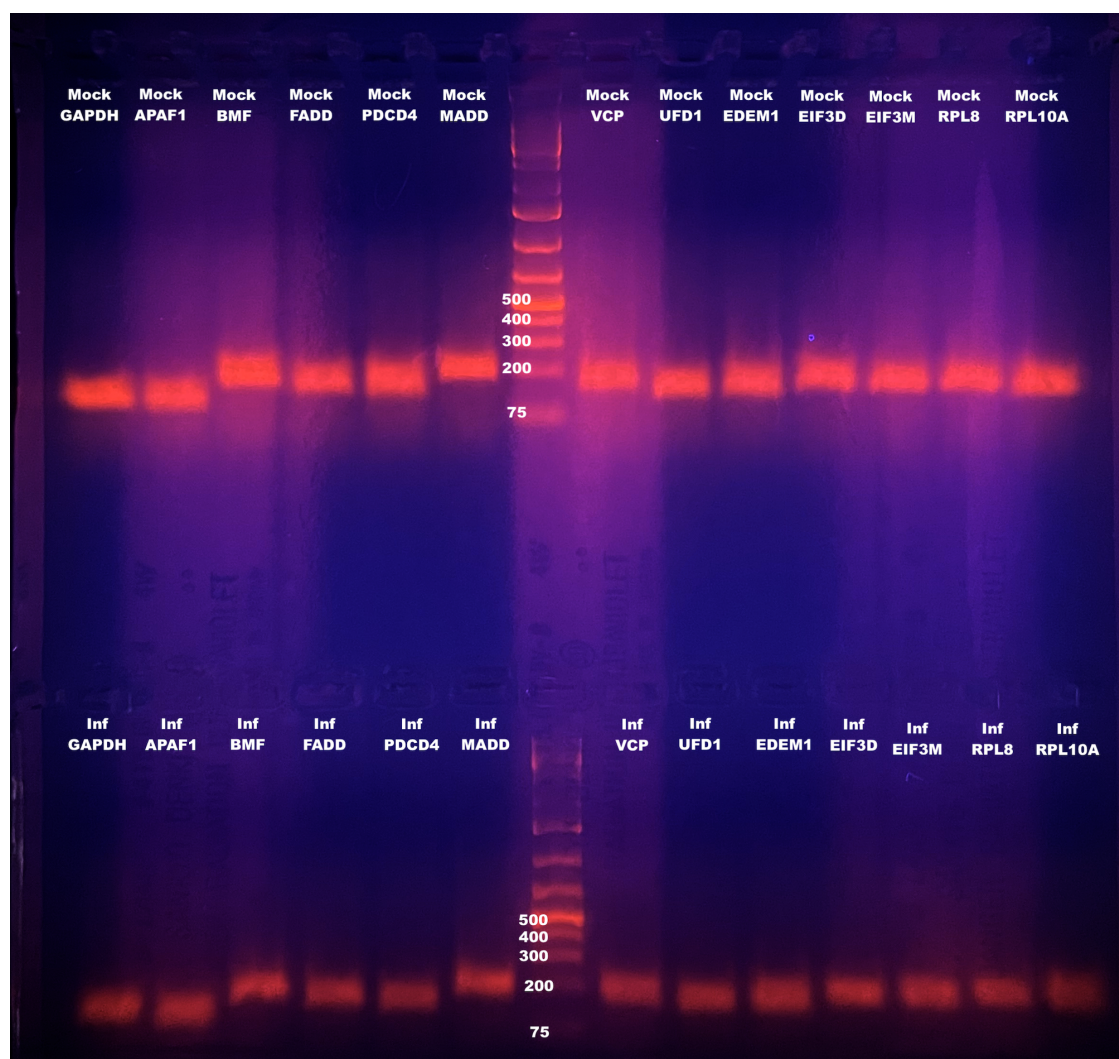


Figure S1: Gel Electrophoresis of RT-qPCR validation reactions. We run a gel electrophoresis of the RT-qPCR reactions in 2% agarose gel to confirm primer specificity. The 13 primer pairs all show excellent target specificity, each amplifying one amplicon of the expected size (see Table S1 for expected amplicon sizes of each primer pair). This was also confirmed in the RT-qPCR melt curves (not shown). Thermo Scientific™ generuler 1 kb plus DNA ladder was used. Mock-infected samples are shown in the top row prefixed with “mock” and infected samples are shown in the bottom row prefixed with “inf”.

Table S1: Primers for RT-qPCR Validation of RNA-seq data

Entrez ID	Target Gene	Forward Primer	Reverse Primer	Amplicon Size
100549497	<i>APAF1</i>	GCTGCGCAAATACCCGAGGTC ^{ExJ}	GCCAGACACAGCATCTGTCACAC ^{ExJ}	133 bp
100550591	<i>BMF</i>	CGGAGACTCTTCTATGGGAATGCTGG ^{ExJ}	CTGCTGATGCCGCTGTATGTGG ^{ExJ}	189 bp
100543065	<i>EDEM1</i>	CTGGACTACAGGTGTTGATAGGAGACG ^{ExJ}	CCACTAACTCTGGCCTCAGTGG	159 bp
100545922	<i>EIF3D</i>	GCACAGAGGAACCTTCGGAGAG ^{ExJ}	GTCACGAGGCTTCTGCTGTGAC ^{ExJ}	180 bp
100545633	<i>EIF3M</i>	CTCTCAGACTGCAGCTACTGAGC ^{ExJ}	GTCTGTGCTGAGGTTCCAGTCAG	179 bp
100540536	<i>FADD</i>	GGAGCTCTGCAACTTCCTCATGG	CCTTCATGTCAGGCCACTCATCAG	167 bp
100303685	<i>GAPDH</i> ^{HK}	CACTATCTTCCAGGAGCGTGACC ^{ExJ}	CTGAGATGATAACACGCTTAGCACCAC	146 bp
100551463	<i>MADD</i>	GAGCTGACGAGGTTGAACTTGCTG ^{ExJ}	CTGGCTCCAATGATAACAAGGTAGTCG	200 bp
100547583	<i>PDCD4</i>	GCACAGTAGAAGTGGAGAATCTGAGTG ^{ExJ}	CTTCCTCAACCGCCTCTTTGC	161 bp
100544053	<i>RPL10A</i>	GGCACCGTCAGGCTGAAGTC ^{ExJ}	GGCATCGTACTTCTTAGCCAGCTTC ^{ExJ}	177 bp
100544011	<i>RPL8</i>	GCCGAGAGACATGGCTACATCAAGG	CAGCTGAGCTTTCTTGCCACAG ^{ExJ}	186 bp
104913522	<i>UFD1</i>	GTGGTCTGCTTCAACATCTGTGGTC ^{ExJ}	GATCTATGAGCTTCGGGTAATGGAGAC ^{ExJ}	154 bp
100548376	<i>VCP</i>	CAAGGCCATAGGAGTGAAGCCTC ^{ExJ}	CTCAGGTTGCTCTCAGACTCACC	171 bp

^{HK}Control (house-keeping) gene^{ExJ}Primer spans exon-exon junction;

469 **Gene symbols:** glyceraldehyde-3-phosphate dehydrogenase (*GAPDH*); apoptotic peptidase activating
470 factor 1 (*APAF1*); Bcl2 modifying factor (*BMF*); FAS-associated protein with death domain (*FADD*); pro-
471 grammed cell death 4 (*PDCD4*); MAP kinase activating death domain (*MADD*); valosin containing protein
472 (*VCP/p97*); Ubiquitin Recognition Factor in ER Associated Degradation 1 (*UFD1*); ER degradation enhanc-
473 ing alpha-mannosidase like protein 1 (*EDEM1*); eukaryotic translation initiation factor 3 subunit D and M
474 (*EIF3D* and, *EIF3M*); ribosomal protein L8 and L10a (*RPL8*, and *RPL10A*).

Table S2: Significantly Enriched KEGG Pathways from DEGs identified at 12- and 24-hpi (Results from the gprofiler2 R package)

Time Point	Regulation	KEGG Term	DEG Count	P-value (Adjusted)
12-hpi	down	Ribosome	35	7.70×10^{-24}
12-hpi	down	DNA replication	11	5.07×10^{-7}
12-hpi	down	Oxidative phosphorylation	19	3.10×10^{-4}
12-hpi	down	Base excision repair	9	1.15×10^{-3}
12-hpi	down	One carbon pool by folate	6	1.27×10^{-3}
12-hpi	down	Mismatch repair	6	3.49×10^{-3}
12-hpi	down	Ribosome biogenesis in eukaryotes	9	1.77×10^{-2}
12-hpi	down	Nucleotide excision repair	8	3.36×10^{-2}
12-hpi	up	Autophagy - animal	13	2.09×10^{-2}
24-hpi	down	Ribosome	41	4.71×10^{-28}
24-hpi	down	Aminoacyl-tRNA biosynthesis	12	3.04×10^{-4}
24-hpi	down	Oxidative phosphorylation	22	4.35×10^{-4}
24-hpi	down	Base excision repair	9	1.15×10^{-2}
24-hpi	down	Carbon metabolism	14	3.14×10^{-2}
24-hpi	down	Propanoate metabolism	6	3.99×10^{-2}
24-hpi	up	Ubiquitin mediated proteolysis	17	7.26×10^{-3}
24-hpi	up	Steroid biosynthesis	5	2.63×10^{-2}

Author Contributions

Conceptualization, Abraham Quaye, Brian D. Poole, Brett E. Pickett, Bradford K. Berges, and Joel S. Griffiths; methodology, Abraham Quaye; software, Abraham Quaye; validation, Abraham Quaye; formal analysis, Abraham Quaye, Brett E. Pickett; investigation, Abraham Quaye; resources, Brian D. Poole; data curation, Abraham Quaye; writing—original draft preparation, Abraham Quaye; writing—review and editing, Abraham Quaye, Brian D. Poole, Brett E. Pickett, Bradford K. Berges, and Joel S. Griffiths; visualization, Abraham Quaye; supervision, Brian D. Poole; project administration, Brian D. Poole; funding acquisition, Brian D. Poole. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The raw sequencing read data (FastQ), gene expression counts, and total DEGs identified at 12- and 24-hpi have been deposited at the NCBI Gene Expression Omnibus (<http://www.ncbi.nlm.nih.gov/geo>) under accession number GSE286211.

All the code/scripts in the entire analysis pipeline are available on GitHub (https://github.com/Abraham-Quaye/host_rna_seq)

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Conflicts of Interest

The authors declare no conflicts of interest.

497 **Abbreviations**

498 The following are abbreviations used in this manuscript:

Abbreviation	Definition
DAVID	Database for Annotation, Visualization and Integrated Discovery
DEG	Differentially Expressed Gene
ER	Endoplasmic Reticulum
ERAD	Endoplasmic Reticulum-associated Degradation
FPKM	Fragments Per Kilobase of transcript per Million mapped reads
GCN	Genome Copy Number
GO	Gene Ontology
HE	Hemorrhagic Enteritis
IMS	Immunosuppression
KEGG	Kyoto Encyclopedia of Genes and Genomes
ORF	Open Reading Frame
RNA-seq	RNA sequencing
RT-qPCR	Reverse Transcriptase Quantitative Polymerase Chain Reaction

Abbreviation	Definition
THEV	Turkey Hemorrhagic Enteritis Virus
UPR	Unfolded Protein Response
VAS	Virginia Avirulent Strain
hpi	Hours Post-infection

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